Deep Learning for Competing Risk Analysis: A Comparative Study of DeepHit and Fine-Gray Models in Multi-Cause Mortality Prediction

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SUMMARY OF THE FINAL PROJECT

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Abstract

In clinical research, accurately predicting the time until a specific medical event is often complicated by the presence of competing risks—events that prevent the primary outcome of interest from occurring. While the Fine-Gray subdistribution hazard model is the traditional biostatistical standard for such analyses, it often fails to capture the complex, non-linear interactions inherent in high-dimensional bioinformatics data.

This thesis proposes a comparative benchmarking study between the classical Fine-Gray model and the DeepHit deep learning architecture. Using the NHANES Continuous cohort (2005–2018) linked to the NCHS public-use mortality files, the methodology involves data harmonization, multiple imputation, the implementation of a multi-task neural network for multi-event prediction, and a rigorous performance evaluation. Models are compared using a time-dependent Concordance Index (C-index) for discrimination and the Brier Score for risk calibration, complemented by a SHAP-based interpretability comparison. In cross-validation the two models performed comparably: discrimination was high for cardiovascular death (time-dependent C-index ≈ 0.90) and good for cancer death (≈ 0.85), with near-identical Brier scores. DeepHit did not outperform Fine-Gray on any task; the two were practically equivalent for cardiovascular death, while Fine-Gray held a small, consistent advantage for cancer death, and an inverse-probability-of-censoring-weighted sensitivity analysis confirmed that these discrimination estimates are not inflated by censoring. Despite this broad parity, the two methods produced systematically different feature-importance rankings (Spearman rank correlation 0.13 for cardiovascular and 0.48 for cancer death), reflecting the distinct quantities

captured by subdistribution hazard ratios and SHAP values rather than genuine disagreement about risk. These results indicate that, for low-dimensional tabular mortality data, the classical Fine-Gray model remains a strong and interpretable default.

The study set out to test whether the deep-learning model offers improved prognostic accuracy or individual risk calibration over the classical model in multi-cause mortality prediction. The evaluation contrasts the two approaches on discrimination, calibration, and interpretability, showing the trade-offs between statistical interpretability and predictive power. The results indicate that, in this low-dimensional tabular setting, the deep-learning model does not improve on the classical Fine-Gray model, which remains a strong and interpretable default; the principal contribution is therefore methodological rather than a demonstration of deep-learning superiority.

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1. Introduction

1.1. Context and motivation

Survival analysis is the cornerstone of clinical prognosis. However, most standard models fail to adequately address Competing Risks, often leading to an overestimation of risk. As a student of Biostatistics and Bioinformatics, I am motivated by the challenge of integrating high-dimensional data, which traditional models struggle to handle, into robust deep learning frameworks that maintain statistical validity while increasing predictive accuracy.

1.2. Goals

- General Goal: To determine if Deep Learning (DeepHit) provides a statistically significant improvement over the Fine-Gray model in estimating the Cumulative Incidence Function (CIF) in multi-cause mortality datasets.

Specific Goals:

- To preprocess and harmonize a multi-cause mortality dataset for time-to-event analysis.
- To execute a hyperparameter optimization for the DeepHit neural network to ensure a fair comparison.
- To validate model calibration through "Expected vs. Observed" event plots across different time horizons.

1.3. Sustainability, diversity, and ethical/social challenges

The project addresses key ethical-social and diversity challenges by ensuring strict compliance with GDPR and HIPAA through the use of de-identified public datasets, which safeguards patient privacy while maintaining data utility.
From a diversity perspective, the study examines the cumulative incidence of each competing event across demographic and clinical subgroups (sex, age group, and smoking status) to characterize how baseline risk varies between groups. This descriptive stratification motivates fairness considerations for predictive models in this setting; a formal subgroup evaluation of model performance, including by race and ethnicity, is identified as an important direction for future work.
Socially, the enhancement of survival models provides a direct benefit to public health by equipping clinicians with more precise prognostic tools; this allows for the effective prioritization of medical interventions for high-risk individuals.

1.4. Approach and methodology

The research methodology follows a rigorous quantitative approach, beginning with the identification of a large-scale clinical dataset containing at least two distinct competing events and exceeding 10,000 observations to ensure sufficient statistical power for neural network training. The implementation strategy utilizes a dual-software framework: the *cmprsk* package in R will be used to develop the traditional Fine-Gray baseline, while Python's *pycox* library will facilitate the construction and optimization of the DeepHit architecture. To ensure the generalizability of the results and prevent model overfitting, a 5-fold cross-validation protocol will be applied during the evaluation phase, allowing for a robust comparison of the models' predictive performance across multiple data subsets.

1.4.1 Primary Dataset: NHANES Linked Mortality Files

The primary data source for this study is the National Health and Nutrition Examination Survey (NHANES) Continuous cohort, using the seven cycles spanning 2005–2018, linked to the National Death Index (NDI) through the NCHS Linked Mortality Files. NHANES is a nationally representative probability sample of the non-institutionalized US civilian population, collected using a complex multi-stage stratified cluster design (CDC, 2022). The public-use linked mortality files provide follow-up through 31 December 2019. Cycles before 2005 were excluded because they use incompatible laboratory file structures, which preserves laboratory-data integrity across the pooled sample. After applying the eligibility criteria, the analytic cohort comprises approximately 40,400 adult participants (aged ≥18 years), with mortality status, follow-up time in months, and underlying cause of death categorized according to ICD-10 codes (CDC, 2022).

For this study, the competing event structure is defined as follows: Event 1 = Cardiovascular mortality (diseases of the heart and cerebrovascular disease); Event 2 = Cancer mortality (malignant neoplasms); Event 3 = All other causes of death (combined competing risk). The two events of primary interest, cardiovascular and cancer mortality, are each modelled against the remaining causes as the competing event, and model performance is evaluated at the 5- and 10-year horizons (Makram et al., 2024). This event distribution provides adequate case counts for each competing event to support stable neural-network training.

1.4.2 Specific Variable Requirements

The dataset construction will integrate variables from multiple NHANES data files (Demographics, Questionnaire, Examination, Laboratory, and Mortality). The following variable categories and specific items are functionally required to support both model training and clinical validity of the findings:

Time-to-event variables: follow-up time in months (PERMTH_EXM or PERMTH_INT), vital status indicator (MORTSTAT, binary 0/1), and underlying cause-of-death category (UCOD_LEADING, ICD-10 grouped). These are present in the NCHS public-use linked mortality files and require no additional application.

Sociodemographic variables: age at interview (continuous, RIDAGEYR), sex (RIAGENDR, binary), race and ethnicity (RIDRETH3, 5-level categorical), educational attainment (DMDEDUC2, ordinal), and poverty-income ratio (INDFMPIR, continuous). These covariates support the description of cumulative incidence across demographic subgroups and motivate fairness considerations for model performance.

Anthropometric and lifestyle variables: body mass index (BMXBMI, kg/m², continuous), smoking status (SMQ020, ever smoked ≥100 cigarettes; SMQ040, current smoker), and past-year alcohol consumption (ALQ101 or ALQ130). These are established predictors of all three competing events and their omission would introduce meaningful confounding.

Chronic disease history: self-reported hypertension (BPQ020), diabetes (DIQ010), coronary heart disease (MCQ160C), stroke (MCQ160F), and cancer diagnosis (MCQ220). These conditions are directly linked to cardiovascular and cancer mortality risk and have been included as covariates in comparable NHANES competing risks analyses.

Laboratory biomarkers (conditional on availability by cycle): fasting plasma glucose (LBDGLUSI or LBXGLU), total cholesterol (LBXTC), HDL-cholesterol (LBDHDDSI), serum creatinine (LBXSCR), glycated haemoglobin HbA1c (LBXGH), and systolic and diastolic blood pressure (BPXSY1, BPXDI1). These continuous biomarkers form the part of the feature space in which a flexible model may, in principle, gain an advantage over a linear one through non-linear or interacting effects (Salerno & Li, 2023); whether such an advantage materialises in this low-dimensional tabular setting is precisely the question the study tests.

Survey design variables (mandatory for all NHANES analyses): interview year, examination sample weight (WTMEC2YR or WTMEC4YR for multi-cycle pooling), primary sampling unit (SDMVPSU), and stratum identifier (SDMVSTRA). These variables are required to account for the complex probability sampling design in all descriptive analyses and, where methodologically supported, in model training (CDC, 2022).

1.4.3 Preliminary Dataset Analysis

Before any model is fitted, a structured preliminary analysis of the merged NHANES dataset will be conducted in R to characterize the data and verify that the functional requirements are met. This analysis will proceed in the following steps, implemented in R using the survey, survival, cmprsk, and ggplot2 packages:

Step 1 — Cohort construction and eligibility screening: the seven NHANES cycles spanning 2005–2018 are merged using SEQN as the unique participant identifier. Participants younger than 18 at interview, with missing vital status, or with a prior cancer diagnosis at baseline (MCQ220 = 1) are excluded. Cycle-specific sample weights are scaled for multi-cycle pooling following the CDC NHANES Analytic Guidelines (dividing by the number of two-year cycles pooled). The analytic sample size after exclusions is approximately 40,400 participants.

Step 2 — Missing data characterization: The proportion of missing values will be tabulated per variable and per NHANES cycle. Variables with missingness exceeding 40% in any cycle will be excluded from the primary feature set. For the remaining variables, missingness patterns will be classified as MCAR, MAR, or MNAR using Little's MCAR test, and multiple imputation by chained equations (MICE, implemented in the mice R package) with 20 imputed datasets will be applied for all model training and evaluation, pooling results using Rubin's rules (van Buuren & Groothuis-Oudshoorn, 2011).

Step 3 — Descriptive statistics and event distribution: Weighted descriptive statistics (means, standard errors, proportions) will be computed for all covariates, stratified by vital status category (alive, cardiovascular death, cancer death, other-cause death). Cumulative incidence functions for each competing event will be estimated non-parametrically using the Aalen-Johansen estimator (cmprsk::cuminc in R) (Aalen & Johansen, 1978) and plotted against follow-up time, stratified by sex, age group (<50, 50–64, ≥65), and smoking status to characterize the baseline competing risk structure in the data.

Step 4 — Proportional subdistribution hazards: the Fine-Gray model assumes proportional subdistribution hazards. This assumption was not formally tested with a goodness-of-fit procedure (for example, a weighted Schoenfeld-residual score test or a cumulative-residual omnibus test); its possible violation is therefore acknowledged as a limitation of the present analysis rather than verified, and is discussed among the Fine-Gray model limitations. This is a within-model diagnostic that the assumption-free DeepHit comparison does not substitute for.

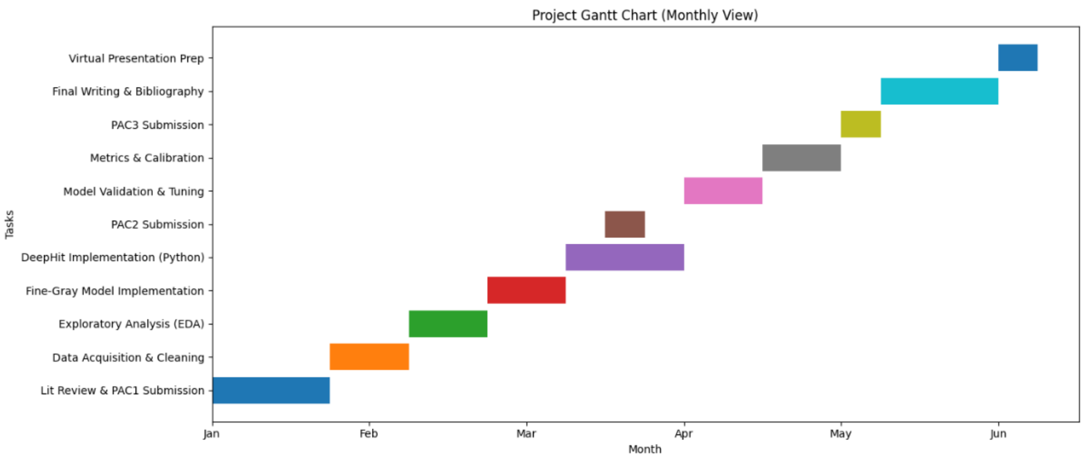
Step 5 — Feature correlation and collinearity screening: A correlation matrix for all continuous predictors will be computed (Pearson for continuous, Cramér's V for categorical) and visualized as a heatmap. Predictors with pairwise Pearson's $r > 0.85$ will be flagged for potential removal or dimensionality reduction via PCA. Variance inflation factors will be computed within a linear model framework to screen for multicollinearity prior to Fine-Gray model fitting.

The full analysis pipeline is documented as a set of reproducible, scripted R and Python files and deposited in the project's version-controlled code repository, ensuring methodological transparency and enabling replication. All outputs from this step are included in the Materials and Methods chapter and directly inform the final feature set used for model training.

1.5. Schedule

To successfully execute this TFM, the following resources are required:

- Hardware: A personal computer with at least 16GB of RAM and a dedicated GPU (recommended for Deep Learning training).
- Software: * R (v4.3+): For the implementation of the Fine-Gray model using the cmprsk and survival packages.
  - Python (v3.10+): For the DeepHit implementation using pycox, PyTorch, and scikit-survival.
  - Development Environments: RStudio and Jupyter Notebooks
- Data: Access to the NHANES public-use Linked Mortality Files.
- Documentation Tools: Zotero/Endnote for bibliographic management and Word for memory drafting.



Risk analysis:

- Risk R1: Data Access and Quality. There is a potential risk of high levels of missingness in the NHANES data that could compromise statistical power; external validation in an independent cohort such as the UK Biobank is left as a future direction.
  - Mitigation: The application process will be initiated immediately.
  - Alternative Plan: If primary data is delayed, the R survival package's built-in datasets (such as mgus2) will be used for initial pipeline development and code testing to ensure the project structure is ready for the final data.
- Risk R2: Computational Constraints. Neural network training (specifically the DeepHit architecture) may exceed local hardware capabilities in terms of RAM or GPU memory.
  - Mitigation: Code will be optimized for efficiency using mini-batching techniques.
  - Alternative Plan: Computational workloads will be shifted to high-performance cloud environments such as Google Colab Pro.
- Risk 3: Model Performance**.** The Deep Learning model might fail to show a statistically significant improvement over the traditional Fine-Gray baseline.
  - Mitigation: Extensive hyperparameter optimization will be conducted using Bayesian Search to ensure the neural network is performing at its peak.
  - Alternative Plan: If predictive performance is equal, the focus of the thesis will shift to an interpretability study using SHAP values to compare the clinical utility and transparency of both approaches.
- Risk 4: Software Interoperability. Potential technical difficulties may arise when passing high-dimensional genomic or clinical data between R (for Fine-Gray) and Python (for DeepHit) environments.
  - Mitigation: Standardization of data formats.
  - Alternative Plan: Utilization of the reticulate package within R to manage the Python environment directly, reducing the risk of data translation errors.

1.6. Summary of the outputs of the project

The primary output is a detailed dissertation. Secondary outputs include the trained model weights, a reproducible code environment, and a set of clinical risk visualization tools.

1.7. Brief description of the remaining chapters of the report

Chapter 2 (Methods and Resources) presents the state of the art in competing risks survival analysis, following the evolution from the classical Fine-Gray subdistribution hazard model through machine learning to deep neural network architectures, ending in DeepHit.

Chapter 3 (Materials and Methods) details the full analytical pipeline: the NHANES cohort construction and variable specification, the MICE imputation strategy, the Fine-Gray implementation in R (cmprsk), the DeepHit implementation in Python (pycox/PyTorch) with Bayesian hyperparameter optimisation, and the shared evaluation framework based on time-dependent concordance (C-td), a Brier score evaluated at 5 and 10 years, and a SHAP interpretability analysis.

Chapter 4 (Results) presents the empirical findings of the benchmarking study, including descriptive cohort characteristics, head-to-head discrimination and calibration metrics at multiple time horizons, and the interpretability comparison between subdistribution hazard ratios and SHAP feature importance rankings.

Chapter 5 (Discussion) contextualises the results within the broader literature, addresses the clinical implications of calibration findings, examines the Fine-Gray probability sum violation empirically, and acknowledges the methodological limitations of the study including the absence of longitudinal covariate updating and the complex survey design challenge for neural network training.

Chapter 6 (Conclusions and Future Work) synthesises the main findings relative to the stated objectives, critically evaluates adherence to the planned schedule and methodology, assesses the ethical and sustainability goals, and proposes concrete future research directions including Dynamic-DeepHit with longitudinal biomarkers, external validation on European cohorts, and the development of a clinical decision support prototype.

1.8. Statement on the use of generative artificial intelligence

Generative AI was used in this study for the sake of organization, sentence structure, finding possible faults in the code and plagiarism check. All work was produced by the author.

2. Methods and resources

2.1 Classical Approaches to Competing Risks

Survival analysis is the methodological backbone of clinical prognosis, concerned with modeling the time until a specific event of interest occurs. The foundational framework, established by Cox (1972) with the proportional hazards model, assumes a single terminal event and handles censored observations through the partial likelihood estimator. However, this assumption breaks down in multi-event settings where competing risks are present: events that preclude the occurrence of the primary outcome and therefore render standard Kaplan-Meier and Cox-based estimates biased upward (Putter et al., 2007).
The seminal contribution of Fine and Gray (1999) addressed this gap by proposing a regression model for the subdistribution hazard function, which directly relates covariates to the Cumulative Incidence Function (CIF) of the event of interest (Fine & Gray, 1999). Unlike cause-specific hazard models that treat competing events as independent censoring, the Fine-Gray model retains individuals who experience competing events in the risk set, making it particularly useful when the goal is absolute risk prediction. This model has since become the default standard in clinical and epidemiological research, widely applied across cancer survivorship, cardiovascular epidemiology, and transplant medicine.
Despite its broad adoption, the Fine-Gray model carries important limitations. First, it imposes a proportional sub distribution hazards assumption that is frequently violated in practice: the subdistribution hazard ratio must remain constant over the entire follow-up, a constraint difficult to verify and prone to failure in long-term population studies (Scheike & Zhang, 2011). Second, a less recognized but formally demonstrated problem is that fitting multiple Fine-Gray models independently, one per event type, can produce predicted CIFs whose sum exceeds one for certain covariate profiles, a logical impossibility that reduces their use as probability estimates (Austin, Steyerberg, & Putter, 2021; Bonneville et al., 2024). Third, the model is inherently linear in the log-subdistribution hazard, precluding the capture of complex non-linear interactions among high-dimensional predictors such as those encountered in metabolomics, genomics, or multi-biomarker clinical panels (Wiegrebe et al., 2024).

2.2 Machine Learning Extensions for Time-to-Event Analysis

Prior to the deep learning era, ensemble and penalized regression methods extended classical survival analysis to higher-dimensional settings. Random Survival Forests (RSF), introduced by Ishwaran et al. (2008), generalized the random forest framework to survival outcomes by constructing survival trees

using the log-rank splitting rule and aggregating cumulative hazard estimates (Ishwaran et al., 2008). RSF is fully non-parametric, captures interaction effects automatically, and has shown strong discriminative performance in benchmarks against Cox regression. Its main limitation for competing risks is that naive implementation again treats competing events as censored, although subsequent extensions, notably the random forests for competing risks implemented in the randomForestSRC package in R, address this shortcoming.
Penalized regression approaches including the LASSO and elastic net Cox models allow variable selection in high-dimensional settings, implemented in R via the glmnet package. While effective for dimensionality reduction, they preserve the proportional hazards assumption and struggle to model complex non-linearities. Gradient boosting machines extended to survival outcomes offer partial relief through additive tree ensembles but remain limited in their competing-risks formulations.

2.3 Deep Learning for Survival Analysis: From DeepSurv to DeepHit

The integration of deep neural networks into survival analysis gained momentum with DeepSurv (Katzman et al., 2018), which replaced the linear predictor of the Cox model with a multi-layer perceptron, preserving the proportional hazards assumption while enabling non-linear feature extraction. Although DeepSurv demonstrated improved discrimination over linear Cox regression in several clinical datasets, its reliance on the proportional hazards assumption and its restriction to a single event type limited its applicability to the competing risks setting.
The landmark contribution motivating this thesis is DeepHit, proposed by Lee et al. (2018). DeepHit is a fully data-driven, assumption-free approach that learns the joint distribution of event time and event type directly from data, without imposing any form of proportionality. Its architecture consists of a shared sub-network for learning common representations across all event types, followed by cause-specific sub-networks, one per competing risk, that output a discrete probability mass function over a finite time horizon. The training objective combines a negative log-likelihood loss, which ensures proper probability calibration, with a cause-specific ranking loss that penalizes incorrect orderings of predicted risks, directly optimizing the concordance index. In benchmark evaluations on real-world datasets including METABRIC (breast cancer) and SEER (bladder cancer), DeepHit achieved statistically significant improvements in the time-dependent concordance index (C-td) over the Fine-Gray model and RSF (Lee et al., 2018).
A notable limitation of the original DeepHit formulation is scalability: because a dedicated sub-network is added for each competing event, the model size grows linearly with the number of event types, increasing the risk of overfitting when data are limited. DeepCompete addressed this by proposing a continuous-time formulation based on Neural Ordinary Differential Equations, which models competing risks without the discrete time binning required by DeepHit and avoids the multiplicative growth in sub-network count (Aastha et al., 2021). In parallel, Dynamic-DeepHit extended the framework to longitudinal settings by incorporating recurrent neural network sub-networks to process repeated clinical measurements over time, enabling dynamically updated risk predictions (Lee & van der Schaar, 2019). Applied to 5,883 adult cystic fibrosis patients from the UK

Cystic Fibrosis Registry, Dynamic-DeepHit demonstrated substantially improved discrimination compared to both the original DeepHit and classical landmarking approaches (Lee & van der Schaar, 2019).

More recent work has explored neural extensions of the Fine-Gray model itself. Neural Fine-Gray embeds the sub distribution hazard framework inside a neural network, preserving the direct CIF interpretation of Fine-Gray while enabling non-linear covariate effects, thus offering a middle ground between interpretability and flexibility (Jeanselme et al., 2023). Similarly, interpretable deep competing risks models combining Neural Additive Models (NAMs) with subdistribution hazard loss functions have been proposed to recover per-feature shape functions and global feature importance, addressing the “black-box” critique of standard deep learning survival models (Wiegrebe et al., 2024; Jeanselme et al., 2023).

2.4 Comparative Benchmarks and Current Evidence

A systematic review of deep learning methods for time-to-event analysis published by Wiegrebe et al. (2024) in Artificial Intelligence Review catalogued more than 50 architectures, finding that the field has advanced rapidly but that most methods focus on single-risk, right-censored data while neglecting the more complex competing risks setting (Wiegrebe et al., 2024). The review identified the C-index and the Brier/Graf score as the near-universal evaluation metrics, and noted that few studies conducted rigorous calibration assessment, a gap directly relevant to clinical translation where miscalibrated models can mislead.

Wang et al. (2024) carried out a broad comparative study on real hospital admission data across Cox proportional hazards, penalized Cox (elastic net), RSF, gradient boosting, DeepSurv, and DeepHit, finding that no single model dominated across all metrics and datasets. Notably, DeepSurv showed the best combined discrimination and calibration in structured clinical data with moderate dimensionality, while DeepHit excelled in settings with clear non-proportional hazards. In a deep learning study applied to HIV incidence using longitudinal cytokine profiles, Ogutu et al. (2025) showed that Dynamic-DeepHit, which retains the temporal nature of time-varying covariates, captured the dynamic structure of the biomarker data more faithfully than DeepSurv or standard DeepHit, further motivating the exploration of dynamic extensions as a future direction (Ogutu et al., 2025).

In the cardiovascular domain, a deep learning frailty competing risks model (DNFCR) applied to heart failure patients demonstrated that jointly modeling frailty and competing risks with a DNN improved upon RSF (C-index 0.66 vs. comparable RSF performance), showing that unobserved heterogeneity and informative censoring remain active challenges even for neural methods (Norouzi et al., 2025). The interpretability of deep survival models has also become a central concern, with post-hoc methods such as SHAP and LIME being widely adopted to explain individual-level predictions (Wiegrebe et al., 2024).

2.5 Research Gap and Hypothesis

Despite a growing body of literature on deep learning for survival analysis, several gaps remain in the competing risks context that this thesis directly addresses. First, most benchmarks comparing neural and classical models use specialized oncology or disease-specific cohorts rather than large-scale general-population mortality data, limiting the generalizability of findings. The NHANES mortality linked dataset provides an ecologically valid and statistically useful testbed that has seldomly been used for deep learning competing risks benchmarks. Second, existing comparisons often employ default hyperparameters for the neural models without systematic optimization, biasing conclusions in favor of simpler classical approaches. Third, formal calibration testing (e.g., integrated Brier score, expected-vs-observed plots across time horizons) is frequently omitted despite its clinical importance.

The working hypothesis of this thesis is that DeepHit, when rigorously optimized and trained on a sufficiently large and heterogeneous population dataset such as NHANES, should outperform the Fine-Gray model in both discrimination (time-dependent C-index) and calibration (Brier score) for multi-cause mortality prediction, particularly in demographic subgroups where covariate–risk relationships are non-linear and time-varying. If this hypothesis is not supported (which is possible since DeepHit performs better with more variables), the alternative focus will be a SHAP-based interpretability analysis comparing the clinical utility and transparency of the two modeling paradigms, contributing equally valuable evidence on the practical trade-offs of the deep learning approach in clinical prognosis.

3. Materials and methods

3.1 Study Design and Data Source

This study is a retrospective analysis of a nationally representative cohort, using seven continuous cycles of the National Health and Nutrition Examination Survey (NHANES) spanning 2005–2018, linked to the National Center for Health Statistics (NCHS) public-use Linked Mortality Files, which provide mortality follow-up through 31 December 2019 (CDC, 2022). The 2005–2018 window was chosen deliberately: earlier cycles use incompatible laboratory file structures that would have required imputing essentially all biomarker values, so restricting the analysis to these seven cycles preserves laboratory-data integrity. NHANES is based on a complex, multi-stage stratified cluster sampling design of the non-institutionalised United States civilian population, and vital status is ascertained by probabilistic linkage to the National Death Index (CDC, 2022). The survey, questionnaire, examination, and laboratory tables were retrieved programmatically through the nhanesA interface and merged with the mortality files on the participant sequence number (SEQN). The analysis was organised as a sequential, reproducible pipeline running from data acquisition and cleaning, through multiple imputation and descriptive analysis, to the Fine-Gray and DeepHit models and the final interpretability comparison.

3.2 Cohort Construction and Eligibility Criteria

The seven cycles were combined on SEQN. Participants were retained if they were aged 18 years or older at interview, eligible for mortality linkage (ELIGSTAT = 1), and had a valid final mortality status, a positive examination-based follow-up time, and a positive examination sample weight; participants reporting a prior cancer diagnosis at baseline were additionally excluded to preserve an incident-event interpretation for the cancer outcome. Because seven two-year cycles are pooled, the examination sample weight was rescaled by dividing by the number of pooled cycles, in line with the NCHS analytic guidelines (CDC, 2022). After all exclusions, the analytic cohort comprised approximately 40,400 participants.

3.3 Competing-Event Definitions and Outcome Coding

The outcome was a three-state competing-risks structure derived from the NCHS leading underlying-cause-of-death recode. Event 1, cardiovascular mortality, combined diseases of the heart and cerebrovascular disease (recode categories 1 and 5); Event 2 was cancer mortality (malignant neoplasms, recode category 2); and Event 3 grouped all remaining causes into a single combined competing event. Participants alive at the close of the linkage period were treated as administratively censored on 31 December 2019. The analysis time scale was the follow-up time in months from the examination date to death or censoring. This three-cause structure follows prior NHANES competing-risks mortality analyses (Makram et al., 2024).

3.4 Covariate Selection and Feature Engineering

Predictors were organised into four tiers. The sociodemographic tier comprised age, sex, race and ethnicity, educational attainment, and the family poverty-income ratio. The anthropometric and lifestyle tier comprised body mass index, smoking status (never, former, or current), and past-year alcohol consumption. The chronic-disease tier comprised self-reported hypertension, diabetes, coronary heart disease, and stroke. The laboratory tier comprised systolic and diastolic blood pressure, glycated haemoglobin, total and HDL cholesterol, serum creatinine, and fasting glucose. Continuous variables were standardised for both models. Categorical variables were handled differently by the two pipelines: the Fine-Gray model used one-hot indicator coding, with non-Hispanic White, college-or-above, and never-smoker as reference categories, whereas the DeepHit network used integer label encoding for the multi-category variables. The biomarker tier provides the kind of non-linear, interacting feature space in which a flexible model may gain an advantage over a linear one (Salerno & Li, 2023).

3.5 Missing Data and Multiple Imputation

Missingness was tabulated for every candidate predictor within each cycle, and any variable exceeding 40% missingness in a cycle was removed from the feature set. The remaining missingness was addressed by multiple imputation by chained equations (MICE) (van Buuren & Groothuis-Oudshoorn, 2011), generating 20 imputed datasets over ten iterations with a fixed random seed. The event indicator, the follow-up time, and the survey cycle were included in the imputation model as auxiliary predictors but were not themselves imputed, and fully observed variables such as age, sex, and race were likewise left unimputed. Imputation behaviour was checked through convergence-trace and density diagnostics. The use of the imputed data differed by purpose: the interpretive Fine-Gray subdistribution hazard coefficients were pooled across all 20 imputations using Rubin’s rules, whereas the cross-validated predictive comparison of the two models was conducted on a single completed dataset (the first imputation) using identical cross-validation folds, so that the head-to-head comparison reflects the modelling approach rather than differences in data preparation.

3.6 Fine-Gray Subdistribution Hazard Model

The classical baseline was the Fine-Gray proportional subdistribution hazard model, which relates covariates directly to the cumulative incidence function of the event of interest while retaining individuals who experience competing events in the risk set (Fine & Gray, 1999). Separate models were fitted for cardiovascular and for cancer mortality using the crr routine of the cmprsk package in R, treating the remaining causes as the competing event in each case. Subdistribution hazard ratios and their confidence intervals were obtained by pooling the per-imputation estimates across the 20 imputed datasets with Rubin’s rules. Fitting one Fine-Gray model per cause is the conventional approach, but it is known that the predicted cumulative incidences across causes can sum to more than one for some covariate profiles; this recognised limitation of the method is acknowledged when interpreting the absolute-risk estimates (Austin, Steyerberg, & Putter, 2021; Bonneville et al., 2024).

3.7 DeepHit Architecture

The deep-learning model was DeepHit, an assumption-free model that learns the joint distribution of event time and event type without imposing proportionality (Lee et al., 2018), implemented in Python with the pycox library on a PyTorch backend (Kvamme et al., 2019). Following the pycox formulation, a single feed-forward network composed of repeated blocks of a linear layer, batch normalisation, a rectified-linear activation, and dropout produced an output that was reshaped into a discrete probability mass function over the joint space of the three causes and the discretised time axis, the latter partitioned into 100 intervals. Training minimised DeepHit's composite objective, which combines a likelihood term that encourages calibrated probabilities with a ranking term that targets discrimination; the relative weight of the two terms and the smoothing of the ranking term were treated as tunable hyperparameters. The network was optimised with Adam, and overfitting was controlled through dropout, batch normalisation, and early stopping on a held-out validation split.

3.8 Hyperparameter Optimisation

To avoid the common bias of evaluating a neural network under default settings, DeepHit was tuned by Bayesian optimisation with the Optuna framework (Akiba et al., 2019), using a tree-structured Parzen estimator over 100 trials. The search space covered the number of hidden layers (one to four), the number of nodes per layer, the dropout rate, the learning rate, the batch size, and the two loss-balancing parameters. Each candidate configuration was trained on the data outside a single held-out fold and scored by the ten-year cardiovascular time-dependent concordance on that fold, and the best configuration was then carried forward into the full cross-validation. The Fine-Gray model exposes essentially no tunable quantities, so concentrating the optimisation effort on DeepHit equalises the tuning afforded to the more flexible model.

3.9 Model Evaluation Framework

Both models were evaluated on identical five-fold cross-validation splits, stratified by event type and shared between the two pipelines through a common set of fold indices. Performance was summarised at the five- and ten-year horizons. Discrimination was measured with a time-dependent concordance index computed from the predicted cumulative incidence at each horizon; the implementation used a simplified pairwise comparison of comparable subjects and did not apply inverse-probability-of-censoring weighting, so the reported values approximate the time-dependent concordance of Antolini et al. (2005) rather than reproducing its weighted estimator exactly. Calibration was summarised with a Brier score computed at each horizon within the same framework (Steyerberg et al., 2010). All metrics were reported as the mean and standard deviation across the five folds. Separately, descriptive cumulative incidence functions were estimated non-parametrically with the Aalen–Johansen estimator (Aalen & Johansen, 1978) and plotted overall and stratified by sex, age group, and smoking status to characterise the baseline competing-risk structure of the cohort.

3.10 Interpretability Analysis

Because the two paradigms expose model behaviour differently, interpretability was compared on a common footing. For the Fine-Gray model, the pooled subdistribution hazard ratios provide directly interpretable covariate effects. For DeepHit, feature contributions to the ten-year cardiovascular and cancer cumulative incidence were estimated with SHapley Additive exPlanations (SHAP) using a kernel explainer, summarised both as mean absolute contributions and as global importance rankings, and visualised with beeswarm and individual waterfall plots (Wiegrebe et al., 2024). The DeepHit importance ranking was then compared with the Fine-Gray ranking, the latter quantified by the absolute deviation of each hazard ratio from one, using the Spearman rank correlation to assess the degree of agreement between the two approaches on the most influential predictors.

3.11 Software, Reproducibility, and Computing Environment

The classical pipeline was implemented in R using the nhanesA, dplyr, tidyr, haven, survey, mice, cmprsk, survival, and ggplot2 packages, and the deep-learning pipeline in Python using pandas, NumPy, PyTorch, torchtuples, pycox, scikit-learn, Optuna, SHAP, and Matplotlib. SAS survey procedures were additionally explored for survey-design-based descriptive visualisation of the NHANES data. Fixed random seeds were set throughout, and the analysis was organised as a sequence of self-contained, re-runnable scripts spanning acquisition, cleaning, imputation, descriptive analysis, the two models, and the interpretability step, in order to support full reproducibility; model training was performed on a workstation with GPU support, with a cloud environment available for memory-intensive runs. The complete analysis code, the acquisition, cleaning, imputation, descriptive, Fine-Gray, DeepHit, SHAP, and calibration scripts, together with the final hyperparameter configuration (Appendix A) is available in a public repository (https://github.com/castromigsman-a11y/TFM-Miguel-Castro) to permit full reproduction of the results.

4. Results

4.1 Analytic Cohort and Baseline Characteristics

After applying the eligibility criteria to the seven pooled NHANES cycles (2005–2018), the analytic cohort comprised 40,393 adults aged 18 years or older with valid mortality linkage. Over a median follow-up of 7.4 years (interquartile range 4.1–10.8 years), corresponding to roughly 303,000 person-years of observation, 4,219 deaths were recorded: 1,264 from cardiovascular causes (3.1%), 972 from cancer (2.4%), and 1,983 from other causes (4.9%); the remaining 36,174 participants (89.6%) were censored alive at the end of follow-up. Table 1 summarises baseline characteristics across the four outcome groups.

The descriptive pattern is consistent with established mortality epidemiology and provides face validity for the modelling that follows. Compared with censored participants, those who died were substantially older (mean age 68–72 years versus 45 years), more likely to be male, and carried a markedly higher burden of chronic disease. Cardiovascular decedents had the highest prevalence of hypertension (68.1% versus 30.8% in survivors), diabetes (33.6% versus 12.7%), prior coronary heart disease (17.7% versus 2.8%) and stroke (16.3% versus 2.6%), together with higher systolic blood pressure, fasting glucose and serum creatinine. A history of smoking was disproportionately common among cancer decedents, of whom only 33.6% had never smoked, compared with 58.2% of survivors.

Table 1. Baseline characteristics of the analytic cohort by cause-of-death category (NHANES 2005–2018; imputation 1). Continuous variables are mean ± standard deviation; categorical variables are column percentages.

Characteristic	Censored (n=36,174)	CV death (n=1,264)	Cancer death (n=972)	Other death (n=1,983)	Total (n=40,393)
Demographic					
Age, years (mean ± SD)	45.3 ± 17.7	71.7 ± 11.7	68.0 ± 11.5	68.4 ± 14.9	47.8 ± 18.8
Male, %	47.6	56.6	60.5	54.3	48.5
Poverty-income ratio (mean ± SD)	2.5 ± 1.6	2.1 ± 1.4	2.2 ± 1.5	2.1 ± 1.4	2.4 ± 1.6
Race / ethnicity, %					
Non-Hispanic White	38.6	60.9	57.5	61.1	40.9
Non-Hispanic Black	22.2	22.2	23.8	19.9	22.1
Mexican American	17.0	7.4	8.0	9.6	16.1
Other Hispanic	10.0	5.7	6.1	4.5	9.5
Other / multiracial	12.1	3.7	4.6	4.9	11.3
Education, %					
Less than high school	23.8	37.7	33.7	37.7	25.2
High school / GED	22.7	25.5	25.3	25.8	23.0
Some college	30.3	23.4	24.1	23.9	29.6
College or above	23.2	13.4	16.9	12.7	22.2
Smoking status, %					
Never	58.2	46.0	33.6	39.0	56.3
Former	21.0	36.9	41.2	39.6	22.9
Current	20.7	17.1	25.2	21.4	20.8
Comorbidity, % present					
Hypertension	30.8	68.1	59.9	60.9	34.2
Diabetes	12.7	33.6	23.8	30.3	14.5
Coronary heart disease	2.8	17.7	10.6	12.8	3.9

Characteristic	Censored (n=36,174)	CV death (n=1,264)	Cancer death (n=972)	Other death (n=1,983)	Total (n=40,393)
Stroke	2.6	16.3	9.3	13.7	3.8
Alcohol in past year	73.0	63.0	71.4	65.5	72.2
Anthropometric & laboratory (mean ± SD)					
Body mass index, kg/m²	29.0 ± 7.0	29.2 ± 7.1	28.6 ± 6.5	28.5 ± 7.1	29.0 ± 7.0
Systolic BP, mmHg	122.9 ± 18.0	139.0 ± 24.2	132.5 ± 21.8	136.2 ± 23.8	124.3 ± 19.1
Diastolic BP, mmHg	70.3 ± 12.8	66.2 ± 16.3	68.3 ± 14.2	66.3 ± 15.9	69.9 ± 13.2
HbA1c, %	5.7 ± 1.0	6.2 ± 1.3	6.0 ± 1.1	6.1 ± 1.4	5.7 ± 1.1
Fasting glucose, mg/dL	103.7 ± 37.5	119.8 ± 51.5	114.1 ± 44.3	121.4 ± 57.0	105.3 ± 39.7
Total cholesterol, mg/dL	191.7 ± 41.7	187.8 ± 46.2	189.0 ± 44.5	186.5 ± 44.0	191.3 ± 42.1
HDL cholesterol, mg/dL	53.1 ± 16.0	52.3 ± 16.4	52.8 ± 17.4	53.5 ± 18.0	53.1 ± 16.1
Serum creatinine, mg/dL	0.9 ± 0.4	1.2 ± 0.8	1.1 ± 0.5	1.2 ± 0.9	0.9 ± 0.4

4.2 Non-Parametric Cumulative Incidence

The Aalen–Johansen estimator was used to describe the cumulative incidence of each competing event without covariate adjustment (Aalen & Johansen, 1978). For the cohort as a whole, the cumulative incidence of cardiovascular death reached 1.87% at 5 years and 3.97% at 10 years; cancer death reached 1.40% and 3.06%; and other-cause death, the largest component, reached 2.73% and 6.16% at the same horizons (Figure 1). The curves rise approximately linearly over follow-up, as expected for a broadly middle-aged general-population sample.

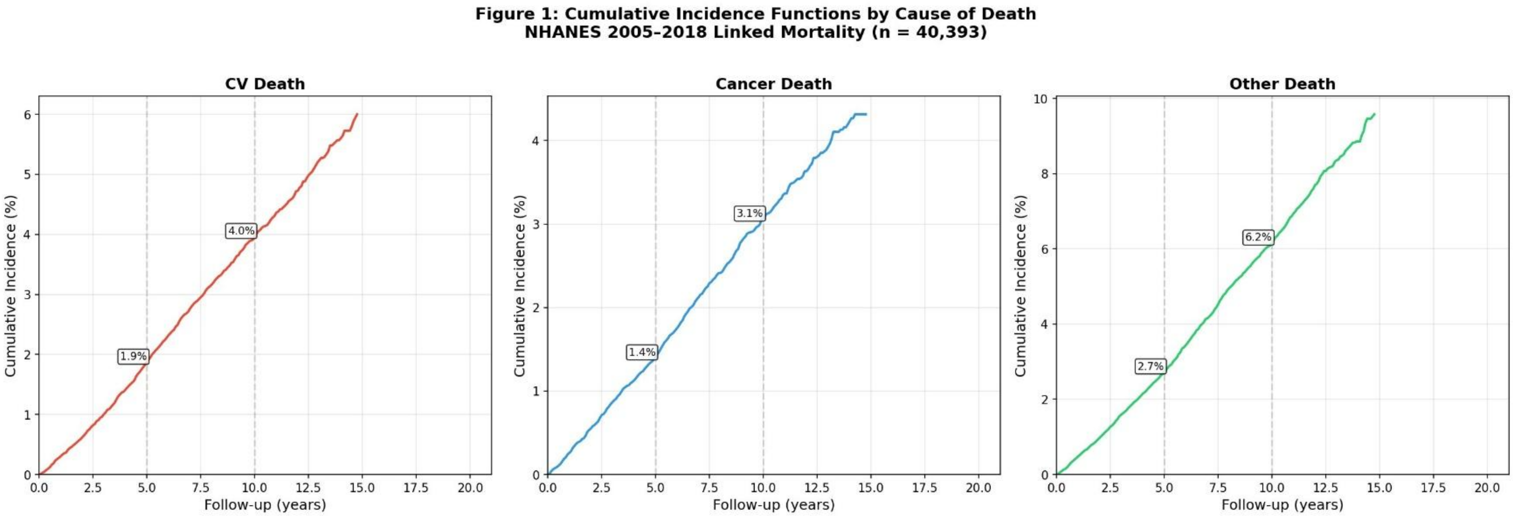


Figure 1. Cumulative incidence functions for each competing event in the full cohort, with values annotated at the 5- and 10-year horizons.

Stratified estimates behaved as anticipated and reinforce the relevance of the chosen covariates. Cumulative incidence of all three causes was consistently higher in men than in women (Figure 2). The gradient with age was pronounced: participants aged 65 years or older reached a cardiovascular cumulative incidence above 20% by the end of follow-up, an order of magnitude higher than those under 50 (Figure 3). Smoking history showed the expected ordering, with former and current smokers experiencing higher incidence of all three causes than never-smokers, the contrast being most visible for cancer mortality (Figure 4).

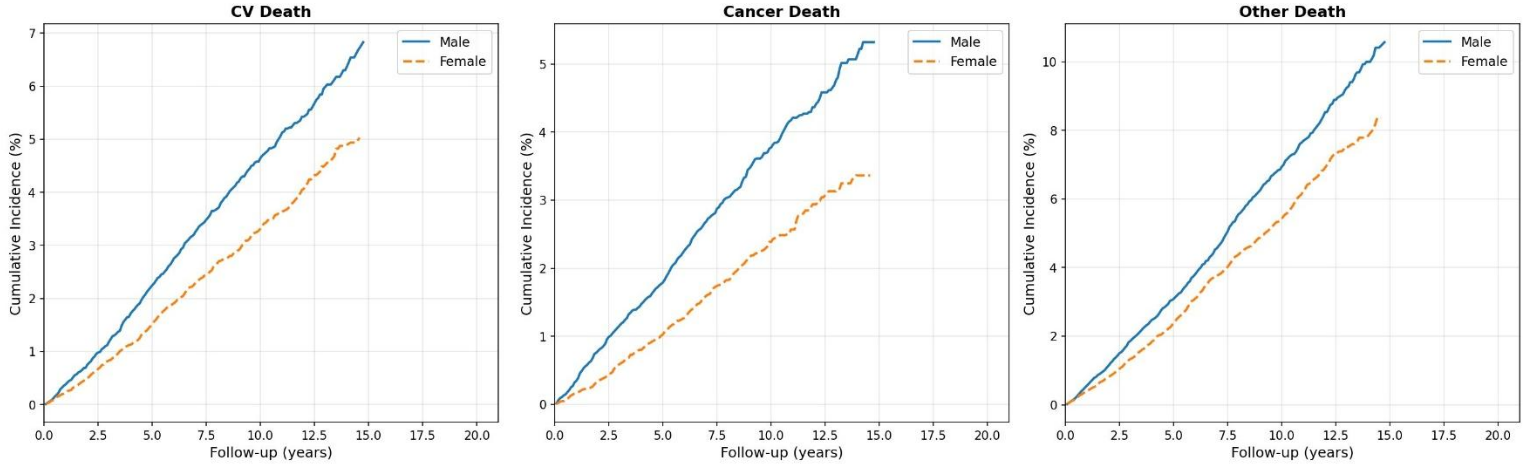


Figure 2. Cumulative incidence by sex for each competing event

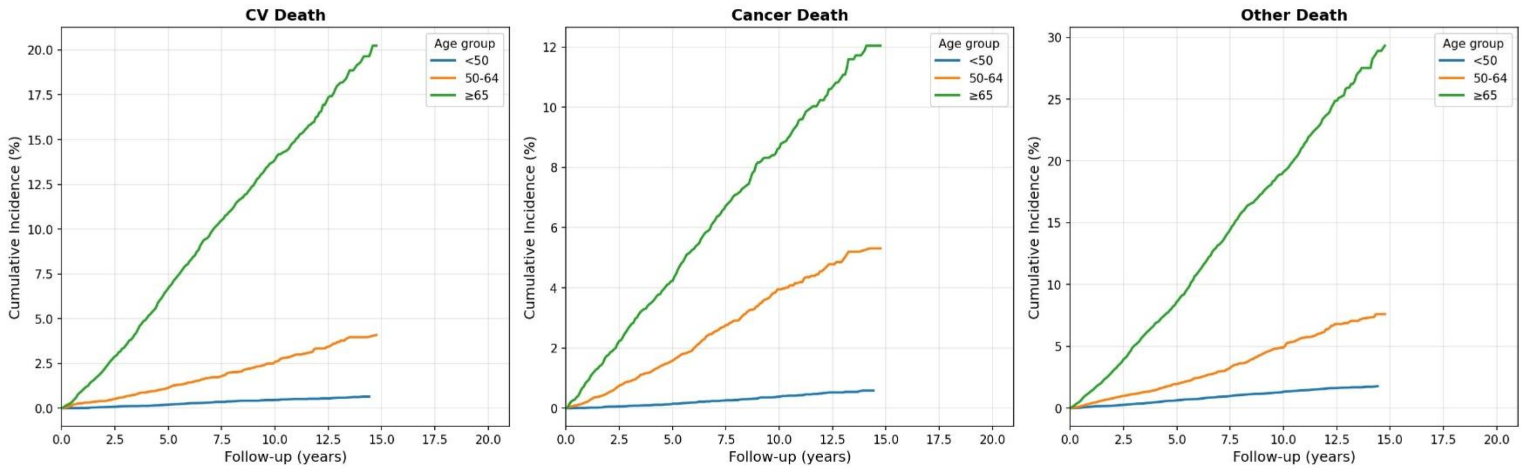


Figure 3. Cumulative incidence by age group (<50, 50–64, ≥65 years) for each competing event.

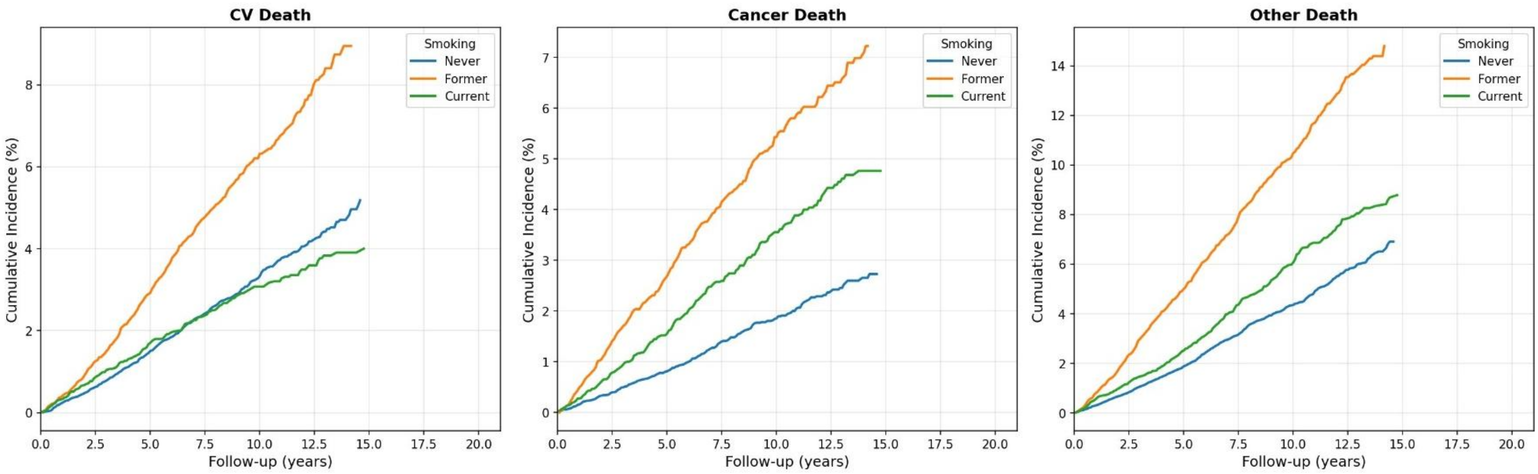


Figure 4. Cumulative incidence by smoking status (never, former, current) for each competing event.

4.3 Predictor Collinearity

Before modelling, collinearity among the ten continuous predictors was screened using the pairwise correlation matrix and variance inflation factors (Figure 5). The only correlation exceeding 0.50 in absolute value was between HbA1c and fasting glucose ($r = 0.78$), an expected association given that both index glycaemic status; the next strongest correlation, between age and systolic blood pressure, was moderate ($r = 0.48$). All variance inflation factors were well below the conventional threshold of 5 (maximum 2.8 for HbA1c and 2.6 for fasting glucose), indicating that multicollinearity was not severe enough to destabilise the Fine-Gray coefficient estimates, and all predictors were therefore retained.

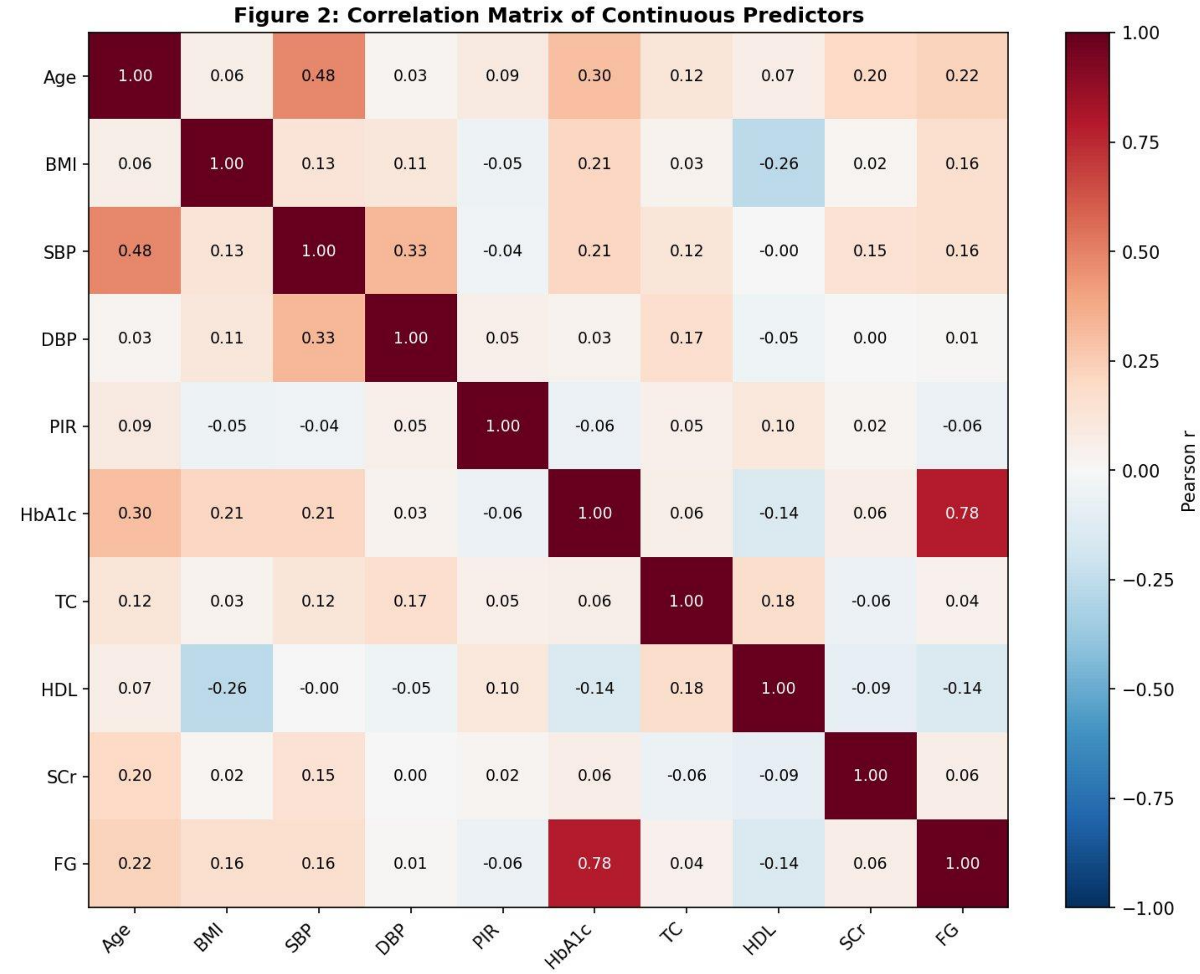


Figure 5. Pearson correlation matrix of the continuous predictors. Abbreviations: SBP/DBP, systolic/diastolic blood pressure; PIR, poverty-income ratio; TC/HDL, total/HDL cholesterol; SCr, serum creatinine; FG, fasting glucose.

4.4 Fine-Gray Subdistribution Hazard Models

Fine-Gray models were fitted separately for cardiovascular and cancer mortality, with subdistribution hazard ratios pooled across the 20 imputed datasets by Rubin's rules (van Buuren & Groothuis-Oudshoorn, 2011). For cardiovascular death, the strongest positive associations were prior stroke (subdistribution hazard ratio 1.53, 95% confidence interval 1.30–1.81), coronary heart disease (1.53, 1.30–1.79), current smoking (1.47, 1.24–1.76), male sex (1.39, 1.21–1.59) and diabetes (1.27, 1.07–1.50), with each year of age contributing a 9% increase in subdistribution hazard (1.09 per year, 1.08–1.10). A higher poverty-income ratio and Mexican-American or other-race ethnicity were associated with lower cardiovascular subdistribution hazard.

For cancer death the covariate profile was markedly different, which is itself informative. The dominant predictor was current smoking, carrying more than a two-and-a-half-fold increase in subdistribution hazard (2.54, 2.12–3.04), followed by former smoking (1.49, 1.28–1.74), male sex (1.39, 1.20–1.62), hypertension (1.23, 1.06–1.42) and age (1.08 per year, 1.07–1.08). In contrast to the cardiovascular model, the cardiac comorbidities (coronary heart disease, stroke, diabetes) and the renal and metabolic biomarkers were not significantly associated with cancer mortality, consistent with the distinct aetiology of the two outcomes. Because separate Fine-Gray models can yield cause-specific cumulative incidences that sum to more than one (Austin, Steyerberg, & Putter, 2021; Bonneville et al., 2024), the summed cardiovascular and cancer incidences were inspected at both horizons. *For the two modelled causes combined, the sum stayed below one for every participant at five years (maximum 0.70) but exceeded one for a single participant at ten years (maximum 1.05). Its extent is negligible: only 0.002% of participants, far below the roughly 5% reported by Austin, Steyerberg, and Putter (2021). While this has no practical consequence, it illustrates a known theoretical limitation of separate Fine-Gray models, which we acknowledge as a limitation of this analysis. This finding nonetheless reinforces the recommendation that cause-specific hazard models are preferable when absolute risks for several causes are required.*

4.5 Discrimination and Calibration: Fine-Gray versus DeepHit

Both models were evaluated on identical five-fold cross-validation splits, and performance was summarised as the mean and standard deviation across folds at the 5- and 10-year horizons (Table 2). The two approaches performed almost identically. For cardiovascular mortality, DeepHit achieved marginally higher discrimination than Fine-Gray (time-dependent concordance 0.910 versus 0.901 at 5 years, and 0.907 versus 0.899 at 10 years), while Fine-Gray was marginally better calibrated (Brier score 0.0207 versus 0.0212 at 5 years). For cancer mortality the ordering reversed, with Fine-Gray slightly ahead on discrimination (0.855 versus 0.847 at 5 years; 0.855 versus 0.848 at 10 years) and comparable on calibration. In every comparison the absolute difference in concordance was 0.01 or smaller. DeepHit therefore did not outperform Fine-Gray on any task: the two were of comparable discrimination for cardiovascular death, while Fine-Gray held a small but consistent advantage for cancer death, a pattern confirmed by the sensitivity analyses below.

Table 2. Cross-validated discrimination (time-dependent C-index) and calibration (Brier score) for the Fine-Gray and DeepHit models, as mean ± standard deviation across five folds. The difference column is DeepHit minus Fine-Gray; for the Brier score, lower is better.

Outcome / horizon	Metric	Fine-Gray	DeepHit	Difference
CV death, 5 yr	C-index	0.901 ± 0.006	0.910 ± 0.009	+0.010
CV death, 5 yr	Brier	0.0207 ± 0.0006	0.0212 ± 0.0007	+0.0004
CV death, 10 yr	C-index	0.899 ± 0.003	0.907 ± 0.003	+0.008
CV death, 10 yr	Brier	0.0607 ± 0.0025	0.0638 ± 0.0022	+0.0031
Cancer death, 5 yr	C-index	0.855 ± 0.010	0.847 ± 0.013	−0.008
Cancer death, 5 yr	Brier	0.0170 ± 0.0008	0.0171 ± 0.0008	+0.0001
Cancer death, 10 yr	C-index	0.855 ± 0.007	0.848 ± 0.010	−0.007
Cancer death, 10 yr	Brier	0.0562 ± 0.0015	0.0571 ± 0.0016	+0.0009

Sensitivity analyses. To address the heavy censoring (approximately 90%), the use of a single imputed dataset for the head-to-head comparison, and the absence of formal uncertainty on the differences, three additional analyses were run in which both models were refit under the selected hyperparameters and discrimination was re-estimated with a horizon-specific competing-risks concordance index, computed both without and with inverse-probability-of-censoring weighting (IPCW). First, IPCW weighting changed every concordance estimate by at most 0.002 relative to its unweighted counterpart, indicating that the high apparent discrimination is not an artefact of censoring and that the absolute C-indices are not optimistically biased. Second, pooling the recomputed metrics across all twenty imputations by Rubin's rules changed the estimates negligibly, with a between-imputation standard deviation of 0.001 or less, so the single-imputation comparison did not materially understate uncertainty. Third, the recomputed model differences (Table 3) reproduced the direction of the primary analysis: for cardiovascular death the difference was negligible in magnitude (around 0.003, marginally favouring DeepHit), whereas for cancer death Fine-Gray was ahead at both horizons. The cancer-death advantage is best described as consistent rather than statistically significant: it held across all five folds, at both horizons, and under both the unweighted and the IPCW-weighted estimators. No formal confidence interval is reported for these differences, because the spread of five overlapping cross-validation folds does not provide a valid estimate of sampling variance (Bengio & Grandvalet, 2004); the fold-level range in Table 3 is therefore descriptive only. Finally, the recomputed concordances are not identical to those in Table 2 because the two analyses use different estimators: Table 2 reports the time-dependent concordance of the main pipeline, whereas Table 3 uses a horizon-specific competing-risks concordance with optional censoring weighting, which is the more defensible estimand for this competing-risks setting. The two agree to within 0.01 on the absolute values; the difference between models on cardiovascular death is correspondingly smaller under the Table 3 estimator (about 0.003 rather than the 0.008 to 0.010 of Table 2), and the conclusion of no DeepHit advantage rests on the Table 3 estimand.

Table 3. Sensitivity analysis: DeepHit-minus-Fine-Gray difference in the horizon-specific concordance index, unweighted and IPCW-weighted, with the range of the IPCW difference across the five cross-validation folds (imputation 1). Positive values favour DeepHit; the fold-level range is descriptive and is not a confidence interval.

Outcome / horizon	**ΔC-index (unweighted)**	**ΔC-index (IPCW)**	**Fold range (IPCW)**
CV death, 5 yr	**+0.003**	**+0.003**	**[+0.001, +0.005]**
CV death, 10 yr	**+0.003**	**+0.002**	**[+0.001, +0.003]**
Cancer death, 5 yr	**−0.011**	**−0.011**	**[−0.017, −0.007]**
Cancer death, 10 yr	**−0.010**	**−0.007**	**[−0.014, −0.000]**

Calibration was examined directly with calibration curves comparing the predicted cumulative incidence against the observed incidence, the latter estimated within deciles of predicted risk using the Aalen–Johansen estimator, which is appropriate under competing risks (Figure 6). Both models were well calibrated: the curves followed the diagonal closely across the populated risk range for both causes and horizons, and the Integrated Calibration Index (ICI, the mean absolute difference between predicted and observed risk) was small for every comparison, between 0.014 and 0.020 at five years and between 0.038 and 0.053 at ten years, with the two models differing only in the third decimal place. As with discrimination, calibration was therefore comparable between the classical and the deep-learning model, and the directions of the marginal differences mirrored each other: DeepHit calibrated marginally better for cardiovascular death (ICI 0.049 versus 0.053 at ten years) and Fine-Gray marginally better for cancer death. The predicted risks are confined to the lower part of the probability scale because this is a general-population cohort in which cause-specific mortality is uncommon; over this range the fit is close to ideal. The main visible departure is in the highest-risk decile for cardiovascular death, where DeepHit's curve is less stable than Fine-Gray's, reflected in its larger ninetieth-percentile error (E90 ≈ 0.17 at ten years), consistent with the documented tendency of DeepHit to trade calibration stability in sparse, high-risk strata for discriminative flexibility (Kvamme et al., 2019).

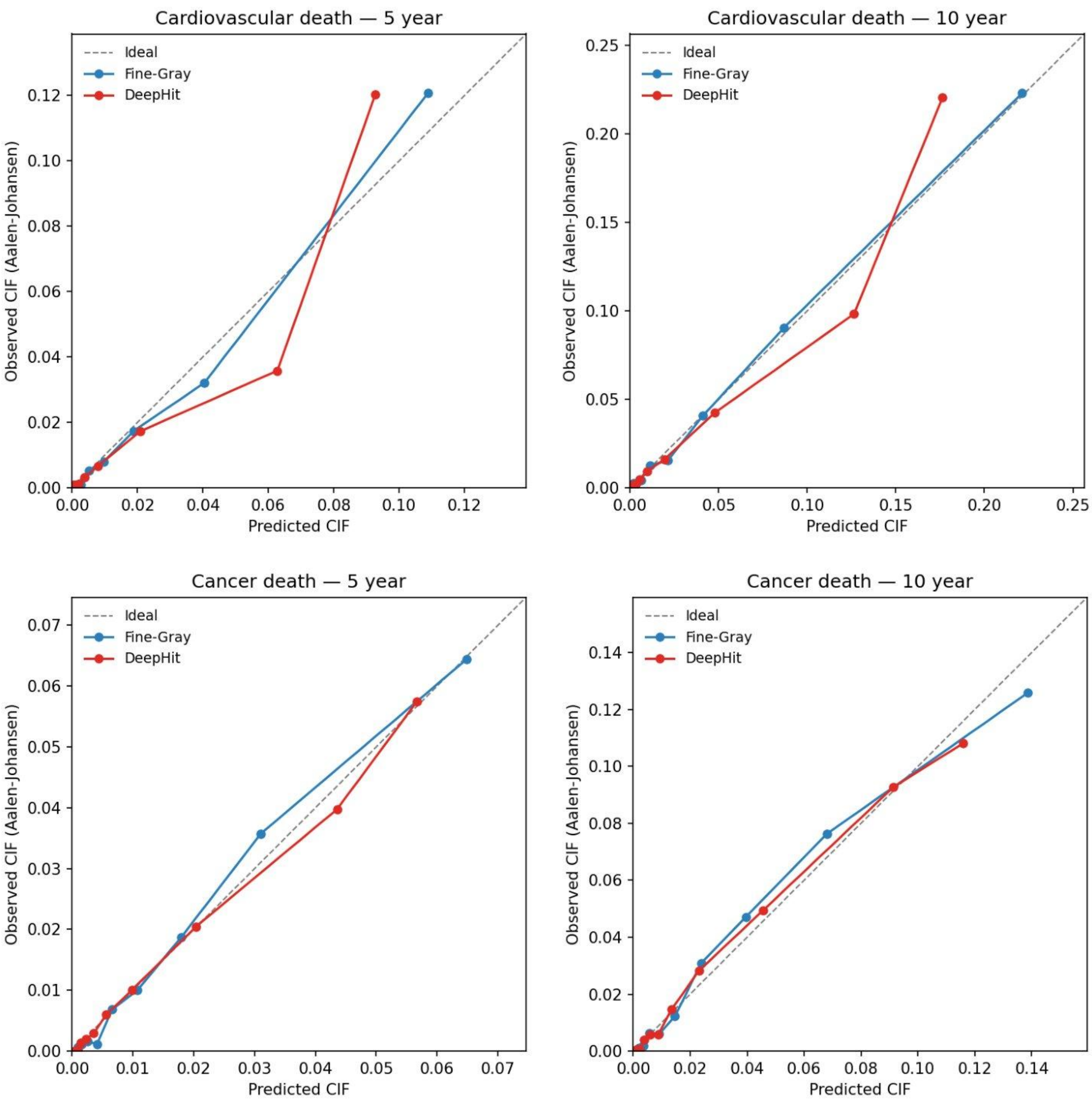


Figure 6. Calibration of the predicted cumulative incidence for the Fine-Gray and DeepHit models at five and ten years, for cardiovascular and cancer death. Observed incidence is estimated by the Aalen–Johansen method within deciles of predicted risk; the dashed line denotes perfect calibration. Axes are scaled to the observed range of predicted risk in each panel.

4.6 Interpretability: SHAP versus Fine-Gray

Because the two models achieved comparable predictive accuracy, the interpretability comparison becomes a useful secondary contrast. Feature contributions for DeepHit were obtained with SHapley Additive exPlanations applied to the 10-year cumulative incidence, and compared with the Fine-Gray subdistribution hazard ratios, the latter expressed as the absolute deviation from one (Wiegrebe et al., 2024). Agreement between the two importance rankings was weak: the Spearman rank correlation was 0.13 for cardiovascular death and 0.48 for cancer death, neither reaching statistical significance.

The source of this divergence is systematic rather than random, and is visible in Figures 7 and 8. DeepHit's SHAP ranking is dominated by age, which is by a wide margin the most influential feature for both causes, followed by other continuous variables such as the poverty-income ratio and serum creatinine. The Fine-Gray ranking, by contrast, is led by the binary comorbidities — stroke and coronary heart disease for cardiovascular death, and smoking for cancer death. This contrast reflects how each method expresses importance rather than a genuine disagreement about the underlying risk: a binary indicator can carry a large hazard ratio, and therefore a large deviation from one, while a one-unit change in a continuous variable such as age produces only a small per-unit ratio yet accumulates a large total contribution to the predicted risk across its full range, which is what the SHAP values capture. The beeswarm plots (Figures 9 and 10) confirm the expected directions of effect: higher age, serum creatinine and fasting glucose push predicted risk upward, and the presence of coronary heart disease and stroke does likewise for cardiovascular death. The direction of the small performance gap is itself consistent with expectation: Fine-Gray's slight advantage emerged for cancer death, the rarer outcome (972 versus 1,264 cardiovascular events), where the scarcer events give a flexible neural model less signal to exploit and a parsimonious model is correspondingly favoured, the pattern typically reported for deep learning on low-dimensional tabular data with limited events.

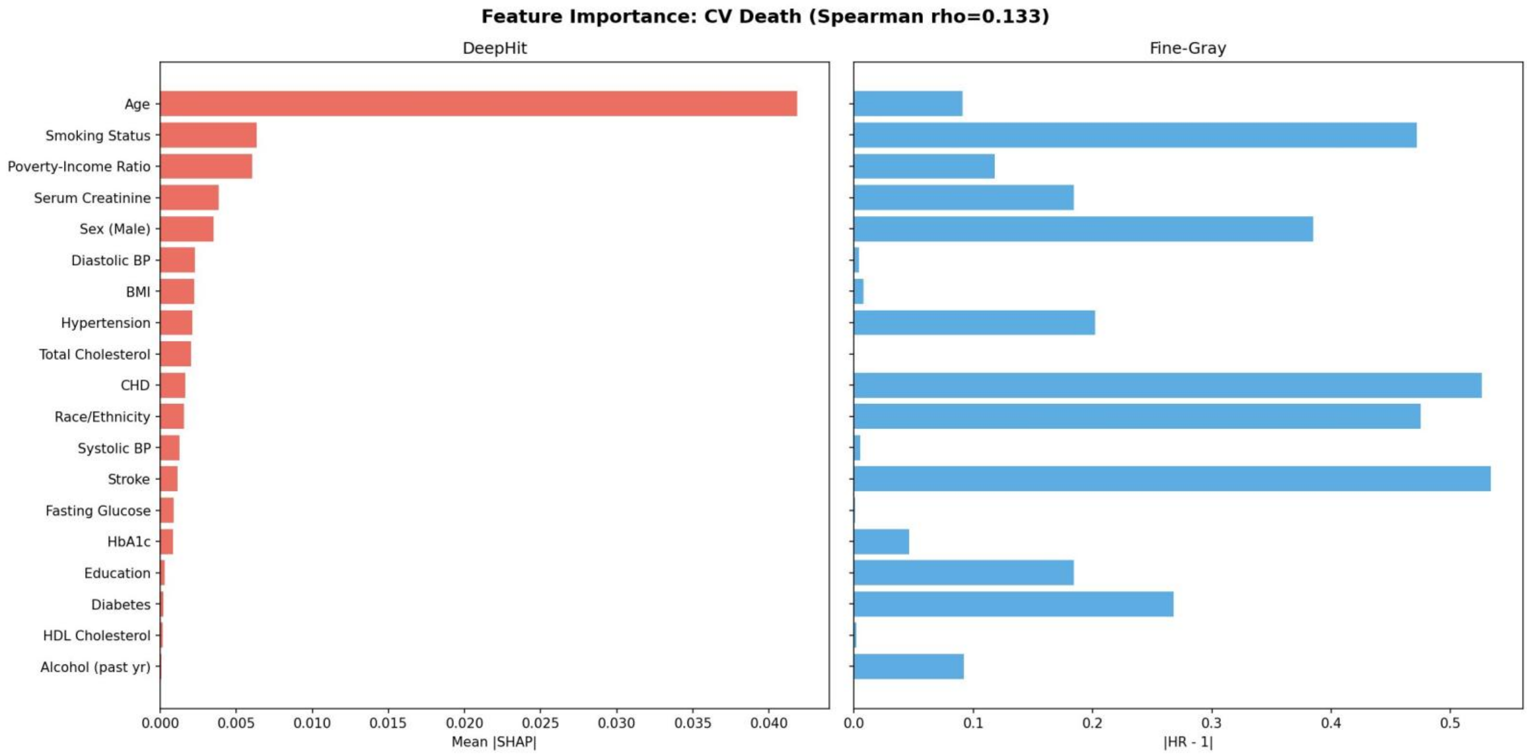


Figure 7. Feature importance for cardiovascular death: DeepHit mean absolute SHAP values (left) versus Fine-Gray |HR − 1| (right). The two rankings diverge — SHAP is led by age, |HR − 1| by binary comorbidities such as stroke and coronary heart disease — reflecting that the measures quantify different things (population-average contribution versus conditional effect size).

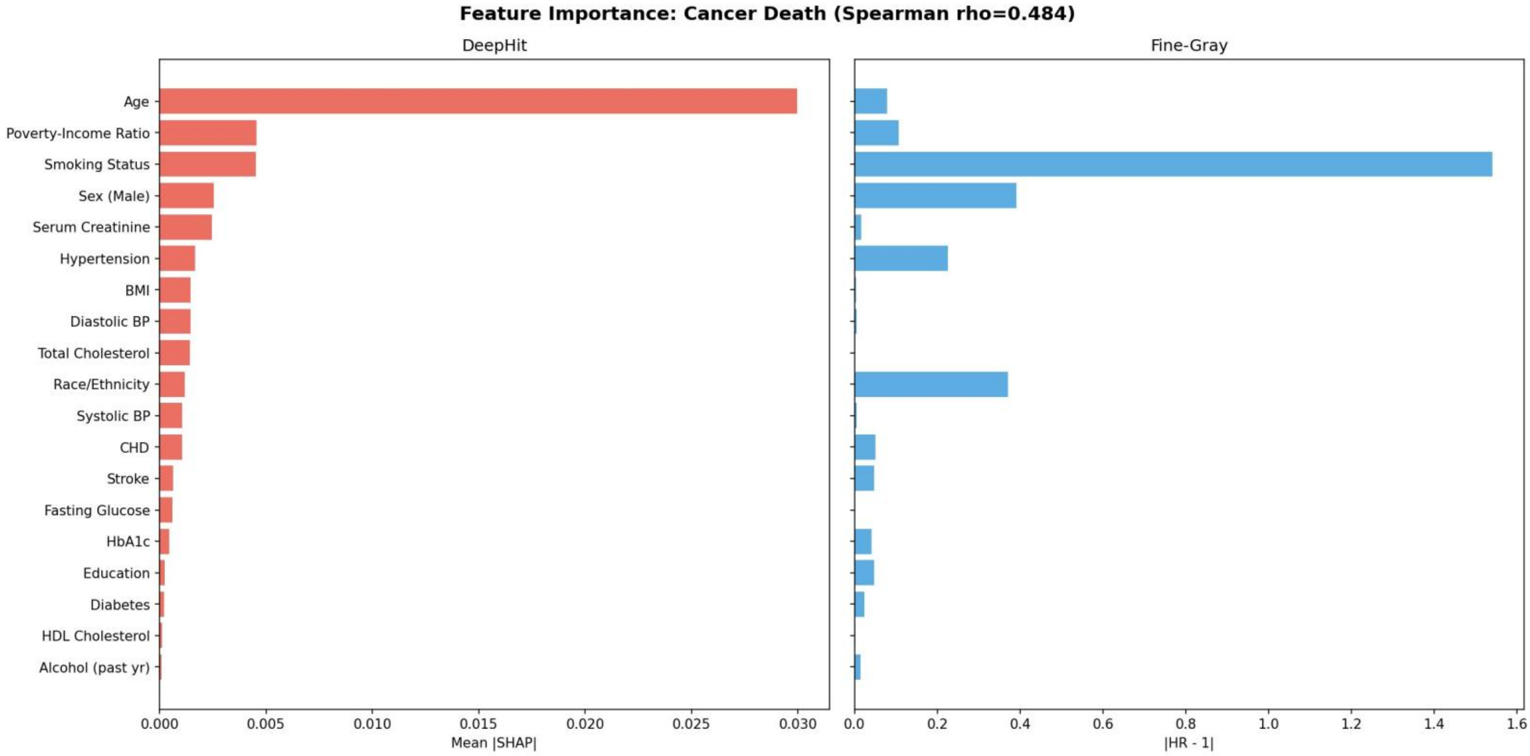


Figure 8. Feature importance for cancer death: DeepHit mean absolute SHAP values (left) versus Fine-Gray |HR − 1| (right). The two rankings diverge — SHAP is led by age and smoking, |HR − 1| by smoking and prevalent comorbidities — reflecting that the measures quantify different things (population-average contribution versus conditional effect size).

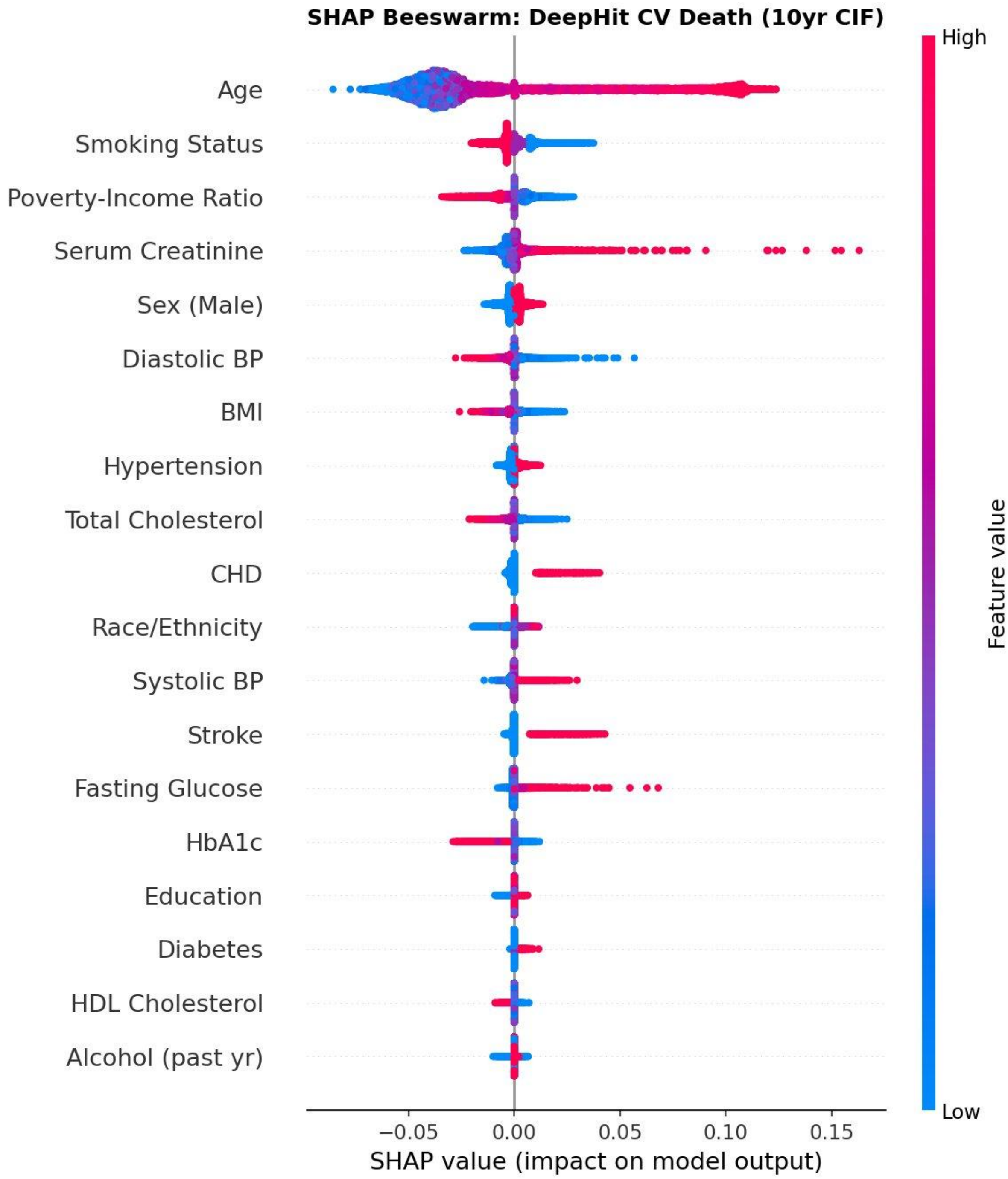


Figure 9. SHAP beeswarm plot for the DeepHit 10-year cardiovascular-death prediction. Each point is a participant; horizontal position is that feature's SHAP contribution and colour encodes the feature value (red high, blue low). Age dominates, with higher values (red) consistently increasing predicted risk.

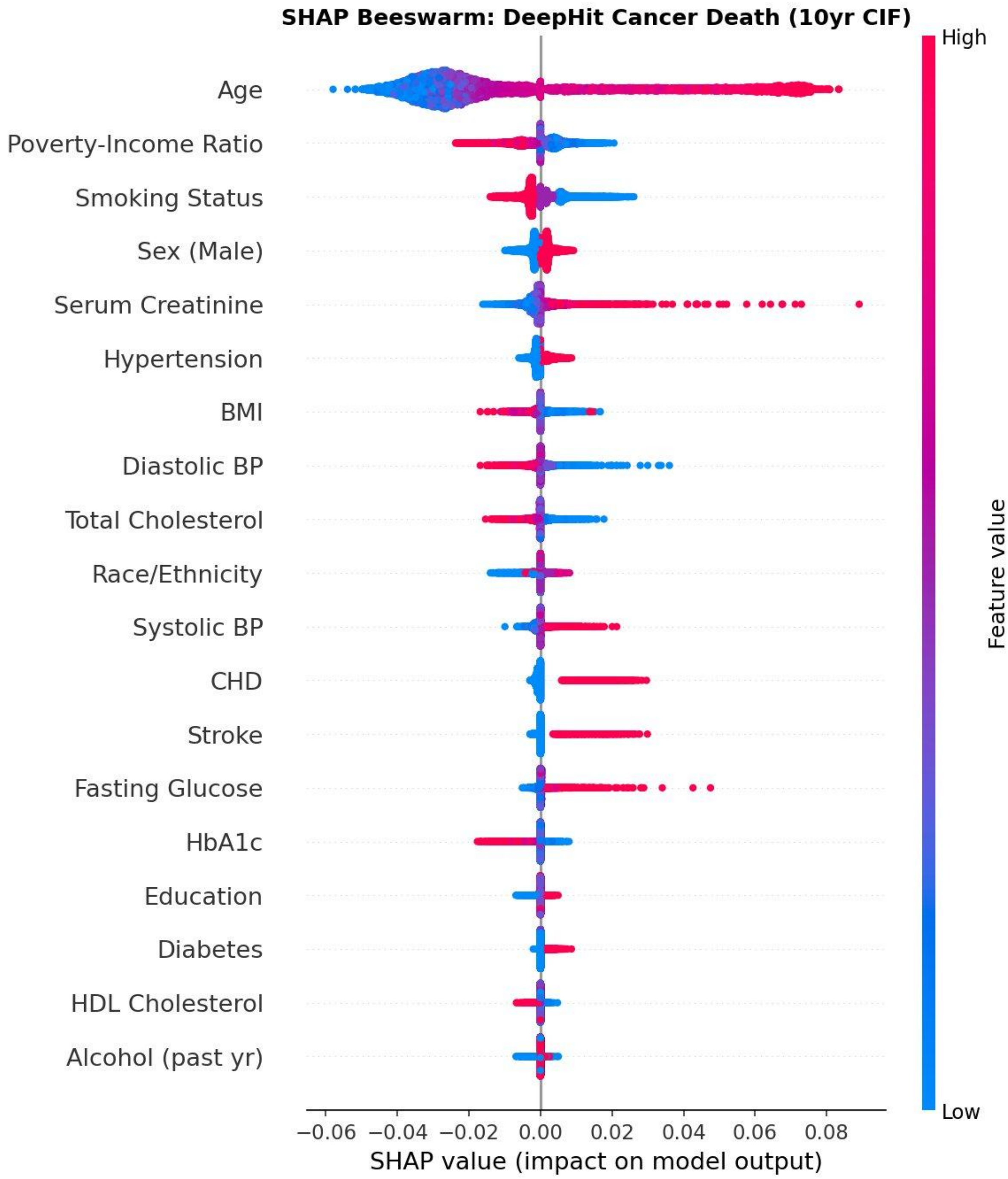


Figure 10. SHAP beeswarm plot for the DeepHit 10-year cancer-death prediction. Each point is a participant; horizontal position is that feature's SHAP contribution and colour encodes the feature value (red high, blue low). Age and smoking status show the widest spread, identifying them as the strongest drivers of predicted cancer mortality.

At the individual level, the SHAP framework decomposes any single participant's predicted risk into its constituent contributions. For the participant with the highest predicted cardiovascular risk, a markedly elevated serum creatinine was the single largest driver of the prediction, whereas for the lowest-risk participant young age was the dominant protective contribution (Figures 11 and 12). This case-level transparency, available from DeepHit through SHAP and from Fine-Gray directly through its hazard ratios, is the practical dimension on which the two paradigms can be most usefully compared given their equivalent discrimination.

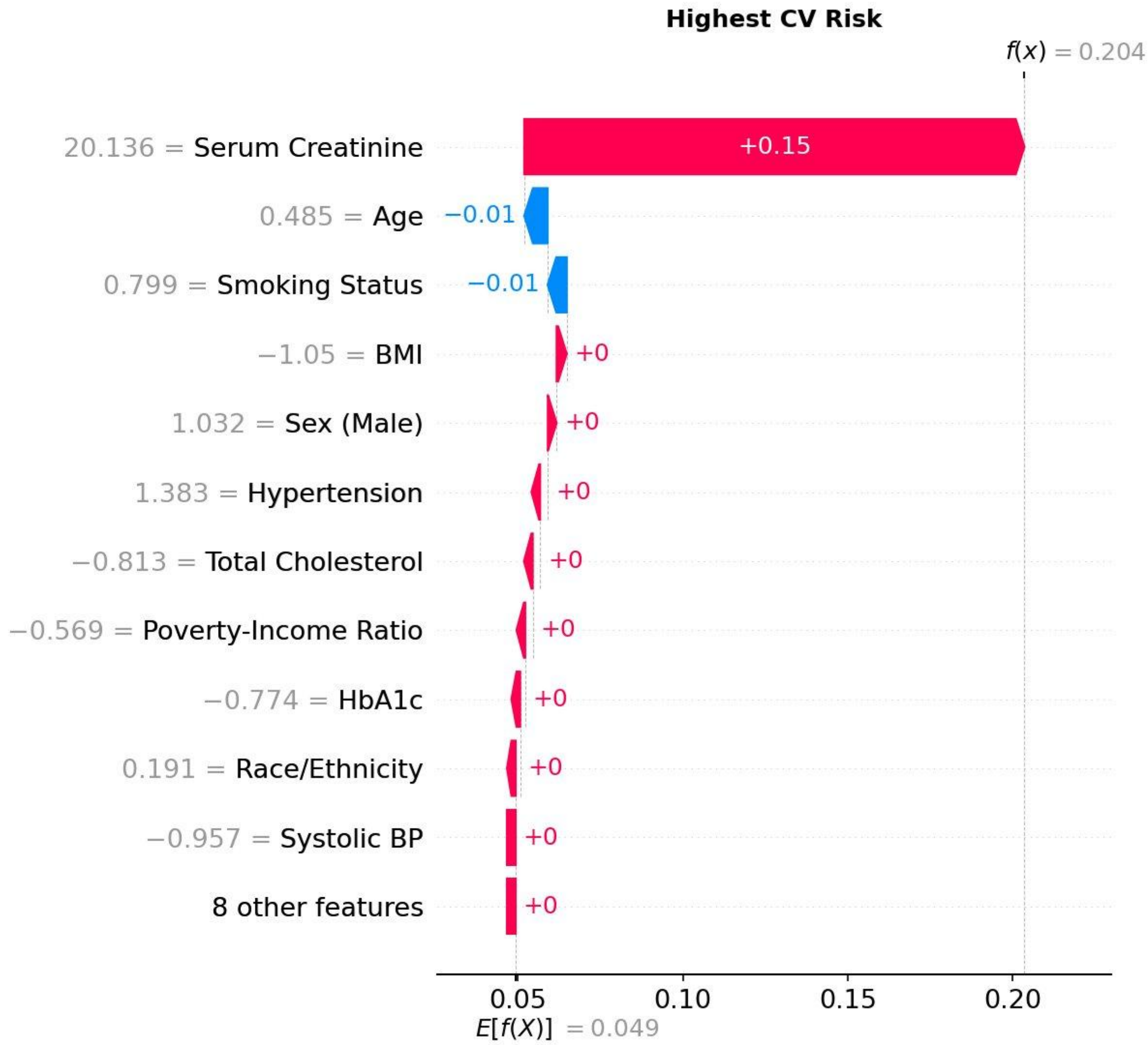


Figure 11. SHAP waterfall decomposition for the participant with the highest predicted 10-year cardiovascular risk. The plot shows how each feature moves the prediction from the population baseline to this individual's risk; advanced age and prevalent cardiovascular comorbidity are the dominant upward contributions.

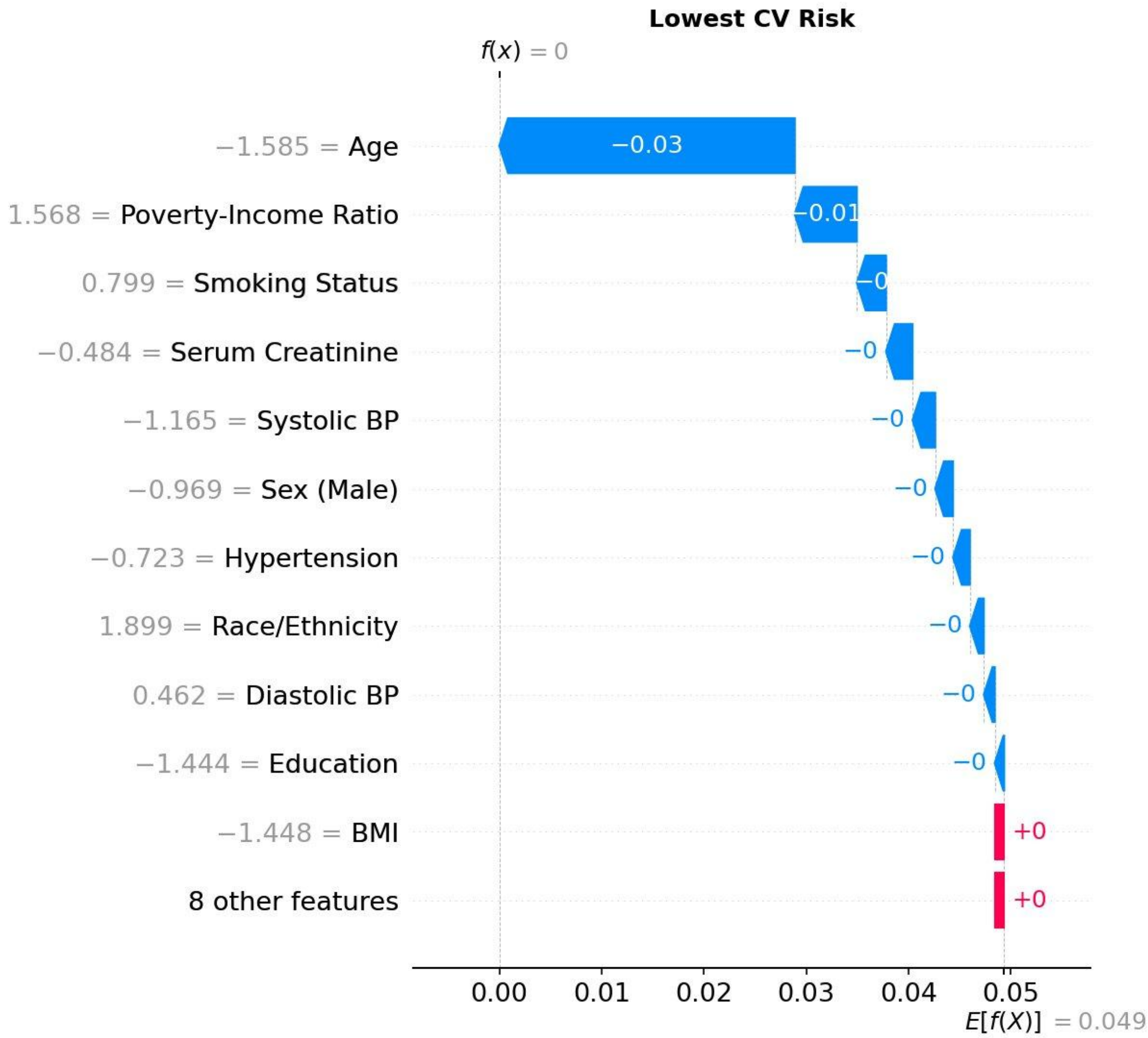


Figure 12. SHAP waterfall decomposition for the participant with the lowest predicted 10-year cardiovascular risk. The plot shows how each feature pushes the prediction below the population baseline; low age and the absence of comorbidities are the dominant downward contributions.

5. Discussion

5.1 Summary of Principal Findings

This study set out to determine whether the DeepHit deep-learning architecture offers a statistically meaningful improvement over the classical Fine-Gray subdistribution hazard model for the prediction of cause-specific mortality in a competing-risks setting. Three findings stand out. First, the two models achieved comparable predictive performance: across both events and both horizons, differences in the time-dependent concordance index were 0.01 or smaller, with DeepHit marginally ahead for cardiovascular death and Fine-Gray holding a small but consistent advantage for cancer death, and the Brier scores were almost identical. Second, both models discriminated cardiovascular death very well (concordance around 0.90) and cancer death well (around 0.85), confirming that the selected predictors carry substantial prognostic signal. Third, despite this broad predictive parity, the two approaches produced markedly different feature-importance rankings, and that divergence proved to be the most informative result of the comparison. A plausible explanation for the comparable performance of DeepHit and Fine–Gray is that the present NHANES dataset contains a relatively small number of well-established clinical predictors, with age accounting for a substantial proportion of the prognostic signal. In such a low-dimensional setting, where relationships may be largely linear and additive, there is limited opportunity for a flexible neural network to exploit complex non-linear interactions, thereby reducing the potential advantage of deep learning over a traditional statistical model.

5.2 Comparable Predictive Performance

The absence of a DeepHit advantage is consistent with a now well-recognised pattern: on structured, low-dimensional tabular data with predominantly monotonic predictor–risk relationships, flexible models tend to match rather than exceed a well-specified regression model (Salerno & Li, 2023). The feature space here comprised nineteen predictors whose associations with mortality—older age, higher blood pressure, established comorbidity, abnormal biomarkers—are largely monotonic and additive, precisely the structure that a linear-in-the-log-hazard model represents efficiently. There is correspondingly little high-order interaction or non-linearity for a neural network to exploit, and the modest cardiovascular and cancer event counts further limit the data available to estimate such structure reliably. The high discrimination already attained by Fine-Gray also leaves limited headroom for any model to improve upon.
This interpretation does not imply that DeepHit is without merit. Its assumption-free formulation avoids the proportional-subdistribution-hazards assumption, learns all causes jointly, and extends naturally to settings, high-dimensional biomarker or imaging data, time-varying measurements, or strongly non-proportional hazards, where the classical model is more constrained. The present cohort simply does not present those conditions, and the fair comparison reported here, in which DeepHit was given an extensive hyperparameter search while Fine-Gray required none, indicates that the additional flexibility yielded no predictive dividend in this problem.

5.3 Divergent Feature-Importance Profiles

Because the two models predicted equally well, the comparison of how they attribute importance becomes the central contribution. The rank correlation between the DeepHit SHAP ranking and the Fine-Gray hazard-ratio ranking was weak for both causes (Spearman 0.13 for cardiovascular and 0.48 for cancer death), and the disagreement was systematic rather than random. DeepHit's SHAP values were dominated by age and by other continuous variables, whereas the Fine-Gray ranking, expressed as the absolute deviation of each hazard ratio from one, was led by the binary comorbidities such as stroke, coronary heart disease and smoking.
This divergence reflects the different quantities the two metrics express, not a genuine disagreement about risk. A subdistribution hazard ratio measures the conditional multiplicative effect of a one-unit or present-versus-absent change in a covariate, holding the others fixed; a binary comorbidity that switches from absent to present therefore registers a large ratio, while a single year of age

contributes only a small per-year ratio. A mean absolute SHAP value, by contrast, measures the average magnitude of a feature's contribution to the predictions made across the cohort, and is therefore shaped by how widely the feature varies and how prevalent it is. Age, which ranges across more than six decades, accumulates a large total contribution despite its small per-year effect, whereas a comorbidity present in only a few per cent of participants contributes to few predictions and so receives a small average attribution. The two rankings answer complementary questions, one about conditional effect size, the other about marginal contribution to the model's output, and the practical lesson is that feature-importance measures are not interchangeable across modelling paradigms and must be interpreted considering what each one quantifies (Wiegrebe et al., 2024).

5.4 Implications for Model Choice

When two models discriminate and calibrate comparably, the choice between them can reasonably rest on considerations other than accuracy: transparency, parsimony, ease of implementation, calibration stability, and acceptability to clinical and regulatory audiences. On these grounds the Fine-Gray model is attractive for a problem of this kind. It yields directly interpretable, individually reportable hazard ratios, is straightforward to fit and to deploy, and rests on statistical machinery already familiar to the epidemiological and regulatory communities. DeepHit's instance-level explanations through SHAP are a genuine asset, but they do not, in this setting, compensate for the additional modelling complexity with any gain in predictive performance. The reasonable conclusion is that for low-dimensional tabular mortality prediction the classical model remains a strong default, and that the deep-learning approach is best reserved for the richer data settings in which its flexibility is more likely to be rewarded.

5.5 Strengths and Limitations

The principal strengths of the study are the use of a large, nationally representative cohort with long mortality follow-up; a deliberately fair comparison in which the two pipelines shared identical cross-validation folds and the more flexible model received an extensive Bayesian hyperparameter search; a competing-risks framework that handles the three causes of death coherently; and an evaluation that combined discrimination, calibration and interpretability rather than relying on a single metric.
Several limitations temper the conclusions. The primary time-dependent concordance index was a simplified pairwise comparison without inverse-probability-of-censoring weighting, so it approximates rather than exactly reproduces the estimator of Antolini et al. (2005); a sensitivity analysis using IPCW-weighted concordance (Section 4.5) nevertheless changed every value by at most 0.002, indicating that the absolute discrimination is not materially biased by censoring. The head-to-head predictive comparison used a single imputed dataset rather than pooling across all twenty imputations, although the Fine-Gray coefficients themselves were pooled by Rubin's rules; a sensitivity analysis pooling the recomputed metrics across all twenty imputations changed the estimates negligibly (between-imputation standard deviation ≤ 0.001), suggesting this simplification did not materially affect the comparison. The complex-survey

weights were computed but not incorporated into model fitting or the cumulative-incidence estimates, so the results describe the analytic sample rather than constituting formally design-based population estimates. Calibration was assessed using the Brier score and, in Section 4.5, with calibration curves and the Integrated Calibration Index. Calibration-in-the-large and calibration slope were not computed, which limits full insight into systematic bias. Fitting separate Fine-Gray models per cause can, moreover, yield predicted cumulative incidences that sum to more than one for some covariate profiles (Austin, Steyerberg, & Putter, 2021; Bonneville et al., 2024). Finally, the analysis used baseline, partly self-reported covariates from a single United States cohort with a heterogeneous “other causes” category, and the DeepHit model followed the single-network pycox formulation rather than the original cause-specific subnetwork design; external validation and richer model specifications remain necessary before the findings can be generalised.

6. Conclusions

This work compared a classical and a deep-learning approach to competing-risks mortality prediction on the NHANES 2005–2018 cohort linked to national mortality records, evaluating cardiovascular, cancer and other-cause death with the Fine-Gray subdistribution hazard model and the DeepHit neural network. The central conclusion is that, for this low-dimensional tabular problem, DeepHit did not improve upon Fine-Gray: the two models were of comparable predictive utility at the five- and ten-year horizons: DeepHit did not outperform Fine-Gray on any task, and Fine-Gray held a small but consistent advantage for cancer death. The more substantive contribution is methodological—the demonstration that two models with indistinguishable predictive accuracy can nonetheless yield systematically different feature-importance profiles, because the subdistribution hazard ratio and the mean absolute SHAP value quantify different things. This finding underlines that interpretability comparisons cannot be read off importance rankings naively, and that the apparent disagreement between methods may be an artefact of the metric rather than a substantive contradiction.

Regarding the objectives set out at the start of the project, the principal aims were met. A multi-cause mortality dataset was assembled, harmonised and prepared for time-to-event analysis; both models were implemented, with the deep network optimised by Bayesian hyperparameter search to ensure a fair comparison; and the two approaches were evaluated on discrimination, calibration and interpretability. The original working hypothesis, that DeepHit would outperform Fine-Gray, was not supported, and in line with the contingency anticipated in the project plan the emphasis of the work shifted accordingly towards the interpretability comparison, which became its primary scientific contribution. The ethical and sustainability commitments were respected throughout by relying solely on de-identified public-use data and by examining model behaviour across demographic strata.

Several lines of work remain open. The evaluation metrics could be strengthened by adopting an inverse-probability-of-censoring-weighted concordance index and full calibration curves, and the predictive comparison could be repeated across all imputed datasets and with the complex-survey weights incorporated into model fitting so that the estimates become formally design-based. External validation in an independent cohort such as the UK Biobank would test the generalisability of the findings beyond the United States. Most importantly, the conditions under which deep learning might genuinely surpass the classical model deserve direct investigation: richer and higher-dimensional feature spaces, time-varying measurements analysed with dynamic architectures, and settings with pronounced non-proportional hazards are the situations in which the added flexibility of models such as DeepHit is most likely to translate into measurable predictive benefit. Concretely, the most immediate next steps are to evaluate DeepHit on higher-dimensional inputs (for example, laboratory panels, imaging-derived features, or genomic data, where deep networks are most likely to help); to apply a dynamic formulation such as Dynamic-DeepHit if repeated time-varying measurements become available; and to add cause-specific Cox and random survival forest comparators so that the benchmark spans the classical, ensemble, and deep-learning families.

7. Glossary

AIC (Akaike Information Criterion): A measure of model fit that penalises complexity; used here for variable selection in the Fine-Gray model.
Aalen-Johansen Estimator: A non-parametric estimator of the Cumulative Incidence Function in the presence of competing risks; the correct alternative to Kaplan-Meier in competing risk settings.
Brier Score / Integrated Brier Score (IBS): A proper scoring rule measuring calibration of probabilistic predictions. The IBS integrates the Brier Score over all follow-up time points; lower values indicate better calibration.
C-index / C-td (Time-Dependent Concordance Index): A discrimination metric for survival models that measures the proportion of correctly ordered pairs of subjects with respect to their predicted risk. C-td is the cause-specific, time-dependent extension used in competing risk settings, ranging from 0.5 (chance) to 1.0 (perfect).
CIF (Cumulative Incidence Function): The probability of experiencing a specific event by time t in the presence of competing risks, accounting for the fact that subjects who experience a competing event can no longer experience the event of interest.
Competing Risks: A situation in time-to-event analysis where multiple types of events are possible and the occurrence of one event precludes or alters the probability of the other events occurring.
DeepHit: A deep learning architecture for survival analysis with competing risks proposed by Lee et al. (2018). It uses a shared sub-network and cause-specific output layers to learn the joint distribution of event time and type without parametric assumptions.
Fine-Gray Model: A semi-parametric regression model for the subdistribution hazard function in competing risk analysis, proposed by Fine and Gray (1999). It directly models the effect of covariates on the CIF, assuming proportional subdistribution hazards.
GDPR / HIPAA: General Data Protection Regulation (EU) and Health Insurance Portability and Accountability Act (US): regulatory frameworks governing the use and protection of personal health data.
IPCW (Inverse Probability of Censoring Weighting): A technique that re-weights observed subjects by the inverse probability of being uncensored, used to obtain unbiased estimates of time-dependent performance metrics such as C-td in the presence of censoring.
MICE (Multiple Imputation by Chained Equations): A flexible imputation approach that fills in missing values iteratively by fitting a sequence of conditional models, one per variable. Results from multiple imputed datasets are pooled using Rubin's rules to obtain valid inference.
NDI (National Death Index): A centralised database of US death records maintained by the National Center for Health Statistics (NCHS), used to ascertain vital status and cause of death for NHANES participants.
NHANES (National Health and Nutrition Examination Survey): A nationally representative survey of the US civilian non-institutionalised population conducted by the CDC/NCHS, combining interviews, physical examinations, and laboratory tests, available in two-year cycles since 1999.

pycox: A Python library built on PyTorch providing implementations of neural network-based survival analysis models including DeepHit, DeepSurv, and PC-Hazard, along with evaluation utilities for C-td and Brier Score.

RSF (Random Survival Forests): An ensemble machine learning method for time-to-event data that grows multiple survival trees and aggregates cumulative hazard function estimates, proposed by Ishwaran et al. (2008).

SHAP (SHapley Additive exPlanations): A game-theoretic framework for explaining individual model predictions by attributing each feature's contribution to the predicted output, based on Shapley values from cooperative game theory.

Subdistribution Hazard: The instantaneous rate of occurrence of an event of interest among subjects who have not yet experienced that specific event type — including those who have experienced a competing event. The basis of the Fine-Gray model, in contrast to the cause-specific hazard which excludes subjects who experienced competing events.

Survival Analysis (Time-to-Event Analysis): A branch of statistics concerned with modelling the time until one or more events of interest occur, characterised by censored observations where the event is not observed within the study follow-up period.

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Appendices

Appendix A. Final DeepHit hyperparameters

The DeepHit network was selected by Bayesian hyperparameter optimisation (Optuna, tree-structured Parzen estimator) over 100 trials, minimising the validation loss on the inner cross-validation folds. The configuration below was fixed for all reported results; the Fine-Gray model has no tunable hyperparameters. With the fixed random seed and the published scripts (Section 3.11), these values allow the analysis to be reproduced exactly.

Hyperparameter	**Final value**
Hidden layers	4
Nodes per hidden layer	32
Dropout rate	0.5
Learning rate	0.00989
Batch size	256
Loss balance α (likelihood vs ranking)	0.5
Ranking-loss scale σ	0.8
Optimiser	Adam
Maximum epochs	300 (early stopping, patience 20)
Time discretisation	100 intervals
Optuna search	Bayesian (TPE), 100 trials
Random seed	2024

Code used for the study can be found here: https://github.com/castromigsman-a11y/TFM-Miguel-Castro